

PHY175: Modern Physics & Electronics

Official Examination Question Bank & Solution Manual

360 Categorized Practice MCQs + Complete Answer Keys & Solutions Manual

LPU Engineering Open Series

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About the Question Bank

This comprehensive Question Bank covers all 6 units of the **PHY175: Modern Physics and Electronics** curriculum. Each unit provides:

- **Section 1: 60 Categorized MCQs** directly reflecting official LPU practice exams across Foundational (Easy), Intermediate (Medium), and Advanced (Hard) tiers.
- **Section 2: Complete Answer Key & Technical Rationales** providing conceptual justifications and numerical working for all 60 MCQs.
- **Section 3: Comprehensive Theory Questions & Worked Solutions** containing step-by-step derivations and analysis.

Contents

1	Unit 1: Solid State Physics	7
1.1	Section 1: Chapter-Wise Multiple-Choice Questions (60 MCQs)	7
1.2	Section 2: Complete Answer Key & Technical Rationales	16
1.3	Section 3: Comprehensive Theory Questions & Worked Solutions	18
2	Unit 2: Fundamentals of Electricity and Devices	19
2.1	Section 1: Chapter-Wise Multiple-Choice Questions (60 MCQs)	19
2.2	Section 2: Complete Answer Key & Technical Rationales	27
2.3	Section 3: Comprehensive Theory Questions & Worked Solutions	28
3	Unit 3: Introduction to Number System and Logic Gates	29
3.1	Section 1: Chapter-Wise Multiple-Choice Questions (60 MCQs)	29
3.2	Section 2: Complete Answer Key & Technical Rationales	37
3.3	Section 3: Comprehensive Theory Questions & Worked Solutions	39
4	Unit 4: Introduction to Combinational Logic Circuits	41
4.1	Section 1: Chapter-Wise Multiple-Choice Questions (60 MCQs)	41
4.2	Section 2: Complete Answer Key & Technical Rationales	49
4.3	Section 3: Comprehensive Theory Questions & Worked Solutions	51
5	Unit 5: Introduction to Sequential Logic Circuits	53
5.1	Section 1: Chapter-Wise Multiple-Choice Questions (60 MCQs)	53
5.2	Section 2: Complete Answer Key & Technical Rationales	61
5.3	Section 3: Comprehensive Theory Questions & Worked Solutions	63
6	Unit 6: Introduction of Arduino and Sensors	65
6.1	Section 1: Chapter-Wise Multiple-Choice Questions (60 MCQs)	65
6.2	Section 2: Complete Answer Key & Technical Rationales	73
6.3	Section 3: Comprehensive Theory Questions & Worked Solutions	75

Chapter 1

Unit 1: Solid State Physics

1.1 Section 1: Chapter-Wise Multiple-Choice Questions (60 MCQs)

Part I: Foundational Level (Questions 1 to 20)

Q1. In the free electron theory of metals, conduction electrons are treated as:

- (a) Positive ions moving through lattice
- (b) Free particles moving through the metal
- (c) Waves confined inside nucleus
- (d) Particles fixed to individual atoms

Q2. Which particles mainly carry electric current in a metal according to free electron theory?

- (a) Bound protons
- (b) Lattice neutrons
- (c) Positive nuclei
- (d) Conduction electrons

Q3. Drift current in a semiconductor is produced mainly by:

- (a) A constant magnetic field
- (b) A carrier concentration gradient
- (c) A uniform temperature alone
- (d) An applied electric field

Q4. Diffusion current occurs when charge carriers move from a region of:

- (a) Low concentration to high concentration
- (b) High concentration to low concentration
- (c) Low potential to high potential only
- (d) Low temperature to high temperature only

Q5. At absolute zero (0 K), the Fermi energy is the energy of the:

- (a) Highest empty electron state
- (b) Lowest forbidden energy state
- (c) Lowest occupied electron state
- (d) Highest occupied electron state

Q6. The Fermi-Dirac distribution function gives the probability that an energy state is:

- (a) Converted into lattice energy
- (b) Occupied by a proton
- (c) Located in a forbidden band

(d) Occupied by an electron

Q7. At the Fermi energy $E = E_F$, the Fermi-Dirac occupation probability is:

- (a) 1
- (b) 0
- (c) $1/4$
- (d) $1/2$

Q8. A range of electron energies that cannot exist in a solid is called a:

- (a) Forbidden energy band
- (b) Valence energy band
- (c) Permitted energy level
- (d) Conduction energy band

Q9. The highest energy band normally occupied by electrons at 0 K in a semiconductor is the:

- (a) Conduction band
- (b) Valence band
- (c) Vacuum band
- (d) Forbidden band

Q10. A hole in a semiconductor behaves like a particle with:

- (a) Negative charge
- (b) Zero charge
- (c) Variable charge
- (d) Positive charge

Q11. Effective mass is used to describe how a charge carrier responds to:

- (a) Nuclear force inside an atom
- (b) An external force in a crystal
- (c) Pressure in a vacuum
- (d) Gravity outside the material

Q12. The Hall effect is the development of a transverse voltage when a current-carrying material is placed in a:

- (a) Longitudinal thermal field
- (b) Uniform gravitational field
- (c) Parallel electric field
- (d) Perpendicular magnetic field

Q13. For a material with one type of carrier having concentration n and charge q , the Hall coefficient is:

- (a) $R_H = q/n$
- (b) $R_H = 1/(nq)$
- (c) $R_H = n/q$
- (d) $R_H = nq$

Q14. A pure semiconductor without intentionally added impurities is called:

- (a) A metallic conductor
- (b) An extrinsic semiconductor
- (c) An ionic insulator
- (d) An intrinsic semiconductor

Q15. Adding a pentavalent impurity to silicon usually produces:

- (a) An intrinsic semiconductor
- (b) An n-type semiconductor
- (c) A perfect insulator
- (d) An p-type semiconductor

Q16. In an intrinsic semiconductor, the Fermi level lies approximately:

- (a) At the bottom of the valence band
- (b) Near the middle of the band gap
- (c) Far above the conduction band
- (d) At the top of the conduction band

Q17. In an n-type semiconductor, the Fermi level shifts closer to the:

- (a) Conduction band
- (b) Bottom of the forbidden band
- (c) Middle of the nucleus
- (d) Valence band

Q18. In a direct band gap semiconductor, an electron can recombine with a hole without requiring a change in:

- (a) Electric charge
- (b) Carrier number
- (c) Crystal momentum
- (d) Electron energy

Q19. Which material is commonly classified as an indirect band gap semiconductor?

- (a) Gallium nitride
- (b) Gallium arsenide
- (c) Silicon
- (d) Indium phosphide

Q20. A solar cell converts light energy directly into electrical energy through the:

- (a) Piezoelectric effect
- (b) Photovoltaic effect
- (c) Thermoelectric effect
- (d) Hall effect

Part II: Intermediate Analytical Level (Questions 21 to 40)

Q21. In the Drude model, $\sigma = \frac{ne^2\tau}{m}$. If electron density doubles while mean collision time halves, conductivity:

- (a) Becomes half as large
- (b) Becomes twice as large
- (c) Remains unchanged
- (d) Becomes four times as large

Q22. Electron concentration increases along $+x$. Which electric-field direction balances electron diffusion current?

- (a) Along the positive x -direction
- (b) No electric field is required
- (c) Perpendicular to x -direction

(d) Along the negative x -direction

Q23. For 3D free electron gas at 0 K, $E_F \propto n^{2/3}$. If density increases 8 times, new Fermi energy is:

- (a) $4E_F$
- (b) $8E_F$
- (c) $16E_F$
- (d) $2E_F$

Q24. Probability that an electronic state at Fermi energy E_F is occupied at nonzero temperature is:

- (a) 1
- (b) 0
- (c) $1/e$
- (d) $1/2$

Q25. At temperature T , a state has $E - E_F = k_B T \ln 3$. What is its Fermi-Dirac occupation probability?

- (a) $1/2$
- (b) $3/4$
- (c) $1/3$
- (d) $1/4$

Q26. When N identical isolated atoms form a crystal, one atomic energy level splits into:

- (a) Exactly two levels
- (b) About N closely spaced levels
- (c) A single level
- (d) A forbidden gap

Q27. Why does a completely filled energy band produce no net electrical current under weak field?

- (a) Contains only positive carriers
- (b) Electrons have no kinetic energy
- (c) All enter conduction band
- (d) Contributions from occupied states cancel across the band

Q28. Near the top of valence band, $d^2E/dk^2 < 0$. What is the standard description of transport?

- (a) Negative holes, negative mass
- (b) Neutral carriers, zero mass
- (c) Free electrons, infinite mass
- (d) Positive holes with positive effective mass

Q29. An electron occupies a band region where $m^* < 0$. If an external force acts along $+x$, its acceleration is:

- (a) Always zero
- (b) Along $+x$
- (c) Along $-x$
- (d) Perpendicular to $+x$

Q30. Current is along $+x$, magnetic field along $+z$. If electrons are majority carriers, where do they accumulate?

- (a) On the negative y side

- (b) On the positive y side
- (c) On the negative z side
- (d) On the positive x side

Q31. A semiconductor slab has $n = 5.0 \times 10^{22} \text{ m}^{-3}$, $t = 0.50 \text{ mm}$, $I = 2.0 \text{ mA}$, $B = 0.50 \text{ T}$. Magnitude of Hall voltage V_H is:

- (a) 0.50 mV
- (b) 0.25 mV
- (c) 2.50 mV
- (d) 1.00 mV

Q32. Hall coefficient is $R_H = -3.9 \times 10^{-4} \text{ m}^3/\text{C}$. The carrier type and approximate concentration are:

- (a) Holes, $1.6 \times 10^{22} \text{ m}^{-3}$
- (b) Electrons, $3.9 \times 10^{22} \text{ m}^{-3}$
- (c) Holes, $3.9 \times 10^{22} \text{ m}^{-3}$
- (d) Electrons, $1.6 \times 10^{22} \text{ m}^{-3}$

Q33. In intrinsic semiconductor $n = p = n_i$. If $\mu_e = 3\mu_h$, what fraction of conductivity is due to electrons?

- (a) 1/2
- (b) 1/4
- (c) 3/4
- (d) 2/3

Q34. Silicon is doped with phosphorus. After donor ionization, which carriers and fixed ions are produced?

- (a) Free electrons and positive donor ions
- (b) Free holes and negative donor ions
- (c) Free electrons and negative donor ions
- (d) Free holes and positive donor ions

Q35. Why does conductivity of intrinsic semiconductor increase strongly with temperature?

- (a) More electrons cross bandgap creating pairs
- (b) Forbidden gap fills completely
- (c) Valence electrons per atom increases
- (d) Crystal becomes metal

Q36. For intrinsic semiconductor, $E_i = \frac{E_C + E_V}{2} + \frac{k_B T}{2} \ln(N_V/N_C)$. If $N_C > N_V$, E_i is:

- (a) Slightly below the middle of the band gap
- (b) Slightly above the middle of the band gap
- (c) At conduction-band edge
- (d) At valence-band edge

Q37. At 300 K, electron concentration in n-type Si increases by $100\times$. How far does E_F shift towards E_C ? ($k_B T = 0.0259 \text{ eV}$)

- (a) 0.119 eV
- (b) 0.059 eV
- (c) 0.460 eV
- (d) 0.259 eV

Q38. Why is direct band-gap semiconductor more efficient for light emission than indirect

gap?

- (a) Recombination requires 2 phonons
- (b) Electron-hole recombination conserves momentum without a phonon
- (c) Valence band has larger resistance
- (d) No available conduction states

Q39. In illuminated solar cell under short circuit, primary role of depletion field is to:

- (a) Make carriers move to same contact
- (b) Prevent light entering
- (c) Separate photogenerated electrons and holes
- (d) Create sub-bandgap photons

Q40. Solar cell receives 100 mW of 2.0 eV light. If 80% photons create collected electrons, photocurrent is:

- (a) 40 mA
- (b) 80 mA
- (c) 64 mA
- (d) 20 mA

Part III: Advanced Mastery Level (Questions 41 to 60)

Q41. Metal B has $2\times$ electron density and $3\times$ effective mass of metal A. If $\sigma_A = \sigma_B$, relationship between collision times is:

- (a) $\tau_B = 6\tau_A$
- (b) $\tau_B = \frac{1}{6}\tau_A$
- (c) $\tau_B = \frac{3}{2}\tau_A$
- (d) $\tau_B = \frac{2}{3}\tau_A$

Q42. In nondegenerate semiconductor, $n(x) = n_0 e^{x/L}$. Which electric field makes total electron current density zero?

- (a) $E_x = \frac{kT}{qL}$
- (b) $E_x = \frac{qL}{kT}$
- (c) $E_x = -\frac{kT}{qL}$
- (d) $E_x = -\frac{qL}{kT}$

Q43. Excess minority electrons in p-type Si: $\Delta n(x) = \Delta n_0 e^{-x/L_n}$ for $x > 0$. With no E -field, directions of electron motion and diffusion current are:

- (a) Electrons move $-x$, current $+x$
- (b) Electrons and current both $-x$
- (c) Electrons and current both $+x$
- (d) Electrons move $+x$, current $-x$

Q44. Two 3D free electron systems at $T = 0$: System B has $n_B = 8n_A$ and $m_B^* = 2m_A^*$. What is $E_{F,B}/E_{F,A}$?

- (a) 2
- (b) 8
- (c) 1/2
- (d) 4

Q45. At temp T , state 1 is at $E_F + k_B T \ln 3$ and state 2 is at $E_F - k_B T \ln 3$. Probabilities that state 1 is occupied and state 2 is empty are:

- (a) Both 3/4
- (b) 3/4 and 1/4
- (c) Both 1/4
- (d) 1/4 and 3/4

Q46. A narrow group of states has constant DOS symmetric about chemical potential μ . Integration limits $\gg k_B T$. Thermal excitation satisfies:

- (a) Electrons above $\mu >$ holes below μ
- (b) Holes below $\mu >$ electrons above μ
- (c) Equality only at $T = 0$
- (d) Number of electrons excited above μ equals number of holes below μ

Q47. Crystal has N atoms, each contributing 1 electron from nondegenerate orbital. Assuming spin degeneracy, band theory predicts:

- (a) Contains $N/2$ states, half filled
- (b) Contains $2N$ states, half filled
- (c) Contains $2N$ states, completely filled
- (d) Contains N states, completely filled

Q48. In nearly-free electron model, why does energy gap open at Brillouin-zone boundary?

- (a) Pauli principle shifts states
- (b) Periodic potential couples degenerate forward and backward waves splitting standing wave energies
- (c) Lattice confines electron permanently
- (d) Collisions remove states

Q49. 2D band dispersion: $E(k_x, k_y) = E_0 - \alpha k_x^2 + \beta k_y^2$ with $\alpha, \beta > 0$. Signs of electron effective-mass components are:

- (a) $m_x^* > 0, m_y^* < 0$
- (b) $m_x^* < 0, m_y^* > 0$
- (c) $m_x^* > 0, m_y^* > 0$
- (d) $m_x^* < 0, m_y^* < 0$

Q50. Nearly full valence band has negative electron curvature near max. Why is it represented by holes with positive effective mass?

- (a) Hole carries negative charge
- (b) Removing electron reverses both charge and current contribution
- (c) Removing electron changes curvature
- (d) Hole is a proton

Q51. Rectangular n-type sample carries current I along $+x$ in B field along $+z$, thickness t . Hall voltage $V_H = V(+y) - V(-y)$ is:

- (a) $V_H = \frac{IB}{net}$
- (b) $V_H = \frac{I}{neBt}$
- (c) $V_H = -\frac{IBt}{ne}$
- (d) $V_H = -\frac{IB}{net}$

Q52. Weak-field Hall coefficient for electrons and holes is $R_H = \frac{p\mu_h^2 - n\mu_e^2}{q(p\mu_h + n\mu_e)^2}$. Hall voltage vanishes when:

- (a) $p\mu_e^2 = n\mu_h^2$
- (b) $p\mu_h = n\mu_e$
- (c) $p\mu_h^2 = n\mu_e^2$

(d) $p\mu_e = n\mu_h$

Q53. Compensated semiconductor has $N_D - N_A = 2n_i$. Assuming thermal equilibrium, electron concentration n is:

- (a) $n = 2n_i$
- (b) $n = (1 + \sqrt{2})n_i$
- (c) $n = (\sqrt{2} - 1)n_i$
- (d) $n = \sqrt{2}n_i$

Q54. Two intrinsic materials have same E_g, T . Material B has $N_{C,B} = 4N_{C,A}$ and $N_{V,B} = N_{V,A}/4$. How does $n_{i,B}$ compare to $n_{i,A}$?

- (a) $n_{i,B} = n_{i,A}$
- (b) $n_{i,B} = \frac{1}{2}n_{i,A}$
- (c) $n_{i,B} = 4n_{i,A}$
- (d) $n_{i,B} = 2n_{i,A}$

Q55. Intrinsic semiconductor has $m_h^* = 4m_e^*$. Relative to midgap, where is E_i ?

- (a) Exactly at midgap
- (b) $\frac{1}{2}k_B T \ln 4$ above midgap
- (c) $\frac{3}{4}k_B T \ln 4$ below midgap
- (d) $\frac{3}{4}k_B T \ln 4$ above midgap

Q56. Fully ionized n-type semiconductor ($N_D \gg n_i$). If N_D increases by $100\times$, how does $E_F - E_i$ change?

- (a) Increases by $2k_B T \ln 100$
- (b) Remains unchanged
- (c) Increases by $k_B T \ln 100$
- (d) Decreases by $k_B T \ln 100$

Q57. Two semiconductors have equal E_g , one direct, one indirect. Why direct emits light more efficiently?

- (a) Equal numbers of states
- (b) Conduction electrons have no effective mass
- (c) Photons supply large momentum
- (d) Band-edge electron-hole recombination conserves crystal momentum without a phonon

Q58. Indirect gap has VB max at $k = 0$ and CB min at $k = k_0 \neq 0$. Near-threshold optical absorption requires:

- (a) Two photons absorbed
- (b) Photon alone provides momentum $\hbar k_0$
- (c) Photon and phonon jointly conserve energy and crystal momentum
- (d) Impurity converts gap

Q59. For ideal solar cell with ideality factor 1, $V_{oc} \approx (kT/q) \ln(I_{sc}/I_0)$. At 300 K, by how much does V_{oc} increase when intensity rises $100\times$?

- (a) 0.238 V
- (b) 0.0259 V
- (c) 0.119 V
- (d) 0.0595 V

Q60. Solar cell has $V_{oc} = 0.65$ V, $I_{sc} = 40$ mA, $V_m = 0.52$ V, $I_m = 35$ mA. What is its fill factor?

- (a) 0.70

- (b) 0.58
- (c) 0.91
- (d) 0.80

1.2 Section 2: Complete Answer Key & Technical Rationales

Part A: Questions 1 to 30

Table 1.1: Answer Key with Rationales (Questions 1 to 30)

Q#	Ans	Technical Rationale	Q#	Ans	Technical Rationale
1	B	Drude theory treats valence electrons as a free classical gas.	16	B	Equal electron and hole densities position E_i near midgap.
2	D	Delocalized conduction electrons are the charge carriers in metals.	17	A	Donor electrons populate near E_C , shifting E_F upwards.
3	D	Drift is carrier motion driven by an applied electric field.	18	C	Extrema align at $\vec{k} = 0$, conserving momentum without phonons.
4	B	Diffusion is driven from higher to lower carrier concentration gradient.	19	C	Silicon and Germanium have indirect band gaps.
5	D	At 0 K, E_F is the cutoff energy of highest occupied state.	20	B	Photons create electron-hole pairs separated by p-n junction field.
6	D	Fermi-Dirac $f(E)$ gives electronic state occupancy probability.	21	C	$\sigma \propto n\tau \implies (2)(1/2) = 1$.
7	D	At $E = E_F$, $f(E_F) = 1/(1 + e^0) = 1/2 = 50\%$ at any $T > 0$ K.	22	A	Diffusion drives electrons toward $-x$. Electric field along $+x$ exerts force in $-x$ to create opposing drift.
8	A	The energy gap E_g contains no allowed electronic wave states.	23	A	$E'_F = (8)^{2/3} E_F = 4E_F$.
9	B	At 0 K, the valence band is completely full and conduction band is empty.	24	D	$f(E_F) = 1/(1 + e^0) = 1/2$ at all $T > 0$.
10	D	Absence of an electron behaves as an effective positive charge $+e$.	25	D	$f(E) = 1/(1 + e^{\ln 3}) = 1/(1 + 3) = 1/4$.
11	B	Effective mass accounts for crystal lattice periodic potential response.	26	B	Wavefunction overlap splits N degenerate states into an energy band of N levels.
12	D	Transverse Lorentz force from perpendicular $\vec{B}$ field produces V_H .	27	D	For every electron in state $+k$, there is an electron in $-k$, summing velocity to zero.
13	B	By definition, $R_H = E_y/(J_x B_z) = 1/(nq)$.	28	D	Negative electron curvature is mapped to holes with $q = +e$ and $m_h^* > 0$.
14	D	Undoped pure semiconductor material is intrinsic.	29	C	$a = F/m^*$; with $m^* < 0$, acceleration is opposite to applied force.
15	B	Pentavalent atoms donate extra electrons, forming n-type.	30	A	$\vec{F} = -e(\vec{v}_d \times \vec{B}) = -e((-v_d \hat{x}) \times B \hat{z}) = -e(v_d B \hat{y}) = -ev_d B \hat{y}$ toward $-y$.

Part B: Questions 31 to 60

Table 1.2: Answer Key with Rationales (Questions 31 to 60)

Q#	Ans	Technical Rationale	Q#	Ans	Technical Rationale
31	B	$V_H = \frac{IB}{net} = \frac{(2 \times 10^{-3})(0.5)}{(5 \times 10^{22})(1.6 \times 10^{-19})(0.5 \times 10^{-3})} = 0.25 \text{ mV}.$	46	D	By particle conservation and Fermi-Dirac anti-symmetry about μ .
32	D	Negative sign $\Rightarrow$ electrons; $n = 1/(R_H e) = 1.6 \times 10^{22} \text{ m}^{-3}.$	47	B	With 2 spin states per orbital, band holds $2N$ electrons. N electrons fill half the band (metal).
33	C	$\sigma_e/\sigma_{total} = \mu_e/(\mu_e + \mu_h) = 3/(3+1) = 3/4.$	48	B	Bragg reflection forms standing waves with different electrostatic potential energies.
34	A	Donating an electron leaves fixed positive donor ion N_D^+ and free conduction electron.	49	B	$m_x^* = \hbar^2/(-2\alpha) < 0$ and $m_y^* = \hbar^2/(+2\beta) > 0.$
35	A	Thermal excitation across E_g causes exponential carrier generation $n_i \propto e^{-E_g/2k_B T}.$	50	B	Absence of negative charge electron moving with negative acceleration behaves identically to positive charge with positive mass.
36	A	$\ln(N_V/N_C) < 0$ when $N_C > N_V$, pulling E_i below midgap.	51	D	Electrons accumulate at $-y$, so $V(-y) > V(+y)$, giving negative Hall voltage $V_H = -\frac{IB}{net}.$
37	A	$\Delta E_F = k_B T \ln(100) = 0.0259 \times 4.605 = 0.119 \text{ eV}.$	52	C	$V_H = 0 \iff R_H = 0 \implies p\mu_h^2 = n\mu_e^2.$
38	B	Direct transition is a 1st-order process requiring no phonon lattice interaction.	53	B	$n - n_i^2/n = 2n_i \implies n = (1 + \sqrt{2})n_i.$
39	C	Built-in electric field sweeps electrons to n-side and holes to p-side.	54	A	$N_{C,B}N_{V,B} = N_{C,A}N_{V,A} \implies n_{i,B} = n_{i,A}.$
40	A	Photon rate $= 3.125 \times 10^{17} \text{ s}^{-1} \implies I = 0.8 \times 1.6 \times 10^{-19} \times 3.125 \times 10^{17} = 40 \text{ mA}.$	55	D	$E_i - E_{mid} = \frac{3}{4}k_B T \ln(m_h^*/m_e^*) = \frac{3}{4}k_B T \ln 4$ above midgap.
41	C	$\sigma = \frac{ne^2\tau}{m} \implies \frac{2n_Ae^2\tau_B}{3m_A} = \frac{n_Ae^2\tau_A}{m_A} \implies \tau_B = \frac{3}{2}\tau_A.$	56	C	$E_F - E_i = k_B T \ln(N_D/n_i) \implies$ increases by $k_B T \ln 100.$
42	C	$J_n = qn\mu_n E_x + qD_n \frac{dn}{dx} = 0 \implies E_x = -\frac{kT}{qL}.$	57	D	Direct transitions conserve $\vec{k}$ directly without 2nd-order phonon scattering.
43	D	Electrons diffuse down gradient along $+x$. Conventional current is along $-x$.	58	C	Photon provides energy $\hbar\omega \approx E_g$; phonon supplies crystal momentum $\hbar k_0$.
44	A	$E_F \propto \frac{n^{2/3}}{m^*} \implies \frac{(8)^{2/3}}{2} = \frac{4}{2} = 2.$	59	C	$\Delta V_{oc} = (0.0259 \text{ V}) \ln(100) \approx 0.119 \text{ V}.$
45	C	$f(E_F + \Delta) = \frac{1}{1+e^{\ln 3}} = \frac{1}{4}.$ Empty prob at $E_F - \Delta$ is $1 - f(E_F - \Delta) = f(E_F + \Delta) = 1/4.$	60	A	$FF = \frac{0.52 \times 35}{0.65 \times 40} = 0.70.$

1.3 Section 3: Comprehensive Theory Questions & Worked Solutions

Theory Question 1.1**[University Exam / Analytical]**

State the fundamental assumptions of classical Drude free electron theory and derive $\sigma = \frac{ne^2\tau}{m}$.

Comprehensive Step-by-Step Solution & Derivation:

Drude model treats valence electrons as a non-interacting classical gas colliding with fixed lattice ions with collision interval τ . Under field E , steady state drift velocity is $v_d = -e\tau E/m$, giving current density $J = ne^2\tau E/m = \sigma E$, yielding conductivity $\sigma = \frac{ne^2\tau}{m}$.

Theory Question 1.2**[University Exam / Analytical]**

Derive the mathematical expression for Hall voltage V_H and explain how it determines carrier type.

Comprehensive Step-by-Step Solution & Derivation:

Under magnetic field $B\hat{z}$ and current $I\hat{x}$, Lorentz force drives carriers to $-y$ until electric Hall field E_H balances magnetic force: $qE_H = qv_d B \implies E_H = v_d B$. With $v_d = \frac{I}{nqw}$, Hall voltage is $V_H = E_H w = \frac{IB}{nqt}$. Sign of V_H indicates carrier polarity.

Chapter 2

Unit 2: Fundamentals of Electricity and Devices

2.1 Section 1: Chapter-Wise Multiple-Choice Questions (60 MCQs)

Part I: Foundational Level (Questions 1 to 20)

Q1. According to Ohm's law, what is voltage across $5\ \Omega$ resistor carrying 2 A?

- (a) 7 V
- (b) 2.5 V
- (c) 10 V
- (d) 15 V

Q2. Which law states algebraic sum of currents at a junction is zero?

- (a) Kirchhoff's current law
- (b) Coulomb's law
- (c) Ohm's law
- (d) Kirchhoff's voltage law

Q3. Two equal resistors in series across 12 V. Voltage across each:

- (a) 6 V
- (b) 3 V
- (c) 24 V
- (d) 12 V

Q4. In a series voltage divider, which resistor gets larger voltage drop?

- (a) Grounded resistor
- (b) Smaller resistor
- (c) First resistor
- (d) Larger resistor

Q5. Total current 4 A enters two equal parallel resistors. Current in each:

- (a) 1 A
- (b) 2 A
- (c) 4 A
- (d) 8 A

Q6. In which bias does PN diode conduct significant current?

- (a) Breakdown
- (b) Zero

- (c) Reverse
- (d) Forward

Q7. Depletion region of PN diode under forward bias:

- (a) Narrower
- (b) Wider
- (c) Unchanged
- (d) Conductive metal

Q8. Primary function of diode rectifier:

- (a) Convert DC to AC
- (b) Increase frequency
- (c) Store charge
- (d) Convert AC to DC

Q9. Ideal forward-biased diode behaves as:

- (a) Closed switch
- (b) Capacitor
- (c) Open switch
- (d) Current source

Q10. Three terminals of BJT:

- (a) Emitter, base, collector
- (b) Gate, source, drain
- (c) Input, output, ground
- (d) Anode, cathode, gate

Q11. Which small BJT current controls larger collector current?

- (a) Leakage
- (b) Emitter
- (c) Reverse
- (d) Base current

Q12. Two types of MOS transistors combined in CMOS:

- (a) LED and diode
- (b) JFET and BJT
- (c) NMOS and PMOS
- (d) NPN and PNP

Q13. Major advantage of CMOS circuits:

- (a) Low static power use
- (b) Large size
- (c) Acoustic output
- (d) Mechanical strength

Q14. Which statement describes RAM?

- (a) Usually volatile
- (b) Read-only
- (c) Stores optically
- (d) Uses tape

Q15. Semiconductor memory used in SSD:

- (a) Dynamic RAM

- (b) Register
- (c) Flash memory
- (d) Cache

Q16. Main purpose of AI accelerator chip:

- (a) Route packets
- (b) Speed up AI computations
- (c) Convert sound
- (d) Measure temperature

Q17. Basic role of sensor in IoT:

- (a) Compile software
- (b) Encrypt packet
- (c) Increase voltage
- (d) Detect a physical quantity

Q18. What is a computer network?

- (a) Storage chip
- (b) Power supply
- (c) Group of connected devices
- (d) Single processing unit

Q19. Physical phenomenon confining light in optical fiber:

- (a) Induction
- (b) Resonance
- (c) Photoelectric
- (d) Total internal reflection

Q20. Processor suited for many parallel calculations:

- (a) SSD controller
- (b) Router
- (c) GPU
- (d) CPU

Part II: Intermediate Analytical Level (Questions 21 to 40)

Q21. Source 12 V connected to series 2Ω and 4Ω . Circuit current:

- (a) 6 A
- (b) 2 A
- (c) 1 A
- (d) 3 A

Q22. Currents 5 A, 2 A enter node; 4 A, I leave. Value of I :

- (a) 3 A
- (b) 2 A
- (c) 1 A
- (d) 7 A

Q23. Supply 20 V across series $3\text{ k}\Omega$, $7\text{ k}\Omega$. Voltage across $7\text{ k}\Omega$:

- (a) 14 V
- (b) 17 V
- (c) 10 V

(d) 6 V

Q24. Divider $R_1 = 2 \text{ k}\Omega$, $R_2 = 8 \text{ k}\Omega$. Parallel load on R_2 main effect:

- (a) Source zero
- (b) Output increases
- (c) Current zero
- (d) Output voltage decreases

Q25. Total current 6 A into parallel 3Ω , 6Ω . Current in 3Ω :

- (a) 1 A
- (b) 4 A
- (c) 2 A
- (d) 6 A

Q26. Parallel branches 4Ω , 12Ω . Current in 4Ω is 9 A. Current in 12Ω :

- (a) 1 A
- (b) 3 A
- (c) 27 A
- (d) 9 A

Q27. Depletion region when silicon diode is forward biased:

- (a) Resistance infinite
- (b) Polarity reverses
- (c) Width decreases
- (d) Width increases

Q28. Silicon diode anode 0.8 V, cathode 0.2 V ($V_D = 0.7 \text{ V}$). Behavior:

- (a) Acts as open circuit
- (b) Reverse biased
- (c) Conducts significantly
- (d) Breakdown

Q29. Full-wave bridge rectifier with 50 Hz AC. Output ripple frequency:

- (a) 25 Hz
- (b) 100 Hz
- (c) 200 Hz
- (d) 50 Hz

Q30. Ideal forward-biased diode in diode switch represents:

- (a) Amplifier
- (b) Open circuit
- (c) Current source
- (d) Short circuit

Q31. BJT active region: $I_C = 2 \text{ mA}$, $I_B = 20 \mu\text{A}$. Current gain β :

- (a) 50
- (b) 10
- (c) 100
- (d) 200

Q32. BJT saturation condition:

- (a) Both reverse
- (b) Both forward

- (c) Emitter forward, collector reverse
- (d) Emitter reverse, collector forward

Q33. CMOS inverter with input LOW:

- (a) Both OFF
- (b) PMOS ON, output HIGH
- (c) Both ON
- (d) NMOS ON

Q34. Why CMOS consumes very little static power:

- (a) Breakdown
- (b) Grounded
- (c) Supply path nonconducting in steady states
- (d) No transistors

Q35. Computer temporary working storage during execution:

- (a) ROM
- (b) SSD
- (c) Dynamic RAM
- (d) Optical fiber

Q36. Why SSD is more shock-resistant than HDD:

- (a) Uses no moving mechanical parts
- (b) Loses data without power
- (c) Stores in magnetic domains
- (d) Rotating platter

Q37. Why GPUs/AI accelerators effective for neural networks:

- (a) Sequential instructions
- (b) Perform many similar operations in parallel
- (c) Hardware replacement
- (d) Eliminate memory

Q38. IoT node sends data only when $\Delta T > 2^\circ\text{C}$. Benefit:

- (a) No calibration
- (b) Zero error
- (c) Reduced energy and communication use
- (d) Higher noise

Q39. Deliver data to every host on local subnet:

- (a) Broadcast
- (b) Loopback
- (c) Point-to-point
- (d) Unicast

Q40. Property allowing fiber signal propagation with low loss:

- (a) Total internal reflection
- (b) Induction
- (c) Capacitive coupling
- (d) Thermal expansion

Part III: Advanced Mastery Level (Questions 41 to 60)

Q41. Node V connected to 10 V through 2Ω , GND through 4Ω , 2 A injected into node:

- (a) $V = 8\text{ V}, I = 1\text{ A}$
- (b) $V = 28/3\text{ V}, I = 1/3\text{ A}$
- (c) $V = 20/3\text{ V}, I = 5/3\text{ A}$
- (d) $V = 12\text{ V}, I = -1\text{ A}$

Q42. Source 24 V, $R_1 = 6\text{ k}\Omega$, $R_2 = 3\text{ k}\Omega$. Parallel R_L giving $V_{out} = 6\text{ V}$:

- (a) $R_L = 4\text{ k}\Omega$
- (b) $R_L = 6\text{ k}\Omega$
- (c) $R_L = 12\text{ k}\Omega$
- (d) $R_L = 3\text{ k}\Omega$

Q43. Source 12 A into 2, 3, 6 Ω . 6 Ω removed. Increase in 2 Ω current:

- (a) 2.4 A
- (b) 0.8 A
- (c) 2.0 A
- (d) 1.2 A

Q44. Diode $I \propto I_S e^{V_D/2V_T}$. Ratio $I(330\text{ K})/I(300\text{ K})$ at 0.6 V:

- (a) 1.39
- (b) 8.00
- (c) 2.79
- (d) 5.57

Q45. Diode $V_D = 0.70 + 10I_D$ with 430 Ω across 5.0 V. Operating point:

- (a) $I_D = 9.77\text{ mA}, V_D = 0.798\text{ V}$
- (b) $I_D = 11.36\text{ mA}, V_D = 0.814\text{ V}$
- (c) $I_D = 10.00\text{ mA}, V_D = 0.800\text{ V}$
- (d) $I_D = 9.30\text{ mA}, V_D = 0.793\text{ V}$

Q46. Bridge rectifier 15 V peak, 50 Hz, 1000 μF , 1 k Ω . V_{DC} and ripple:

- (a) $V_{DC} = 13.53\text{ V}, V_r = 0.271\text{ V}$
- (b) $V_{DC} = 13.46\text{ V}, V_r = 0.269\text{ V}$
- (c) $V_{DC} = 13.53\text{ V}, V_r = 0.135\text{ V}$
- (d) $V_{DC} = 14.23\text{ V}, V_r = 0.142\text{ V}$

Q47. Diode logic: 5 V pull-up to Y , diodes from Y to inputs A, B . Function implemented:

- (a) NAND
- (b) OR
- (c) AND function, low input clamping Y near 0.7 V
- (d) NOR

Q48. NPN switch for 120 Ω , 12 V relay ($V_{CE} = 0.2\text{ V}, V_{BE} = 0.8\text{ V}$, driver 3.3 V, $\beta_{forced} = 10$). Base resistor:

- (a) 330 Ω
- (b) 220 Ω
- (c) 270 Ω
- (d) 250 Ω

Q49. Bias: $V_{TH} = 2.0\text{ V}, R_{TH} = 20\text{ k}\Omega, \beta = 99, V_{BE} = 0.7\text{ V}, R_E = 1\text{ k}\Omega$. I_C :

- (a) 0.87 mA

- (b) 1.07 mA
- (c) 1.50 mA
- (d) 1.29 mA

Q50. CMOS $C = 20 \text{ pF}$, $\alpha = 0.25$. Initial 1.2 V, 500 MHz $\rightarrow$ 1.0 V, 400 MHz. New power and percent:

- (a) 2.9 mW, 80.0%
- (b) 2.0 mW, 55.6%
- (c) 2.5 mW, 69.4%
- (d) 2.4 mW, 66.7%

Q51. CMOS NAND gate with $A = 0, B = 1$. Description:

- (a) A-pMOS, B-nMOS ON, LOW
- (b) A-pMOS, B-nMOS ON, output HIGH
- (c) B-pMOS, A-nMOS ON, HIGH
- (d) Both pMOS OFF

Q52. Memory 64-bit word SECDED Hamming. Minimum stored bits:

- (a) 72 bits
- (b) 70 bits
- (c) 73 bits
- (d) 71 bits

Q53. SSD 1.2 TB NAND, 3000 cycles, write amp 2.5. Host bytes:

- (a) 3.00 PB
- (b) 3.60 PB
- (c) 1.20 PB
- (d) 1.44 PB

Q54. Accelerator Peak 100 TOPS, Bandwidth 800 GB/s. Bounds at intensities 60, 150 ops/byte:

- (a) 60, 100 TOPS
- (b) 48, 100 TOPS
- (c) 100, 100 TOPS
- (d) 80, 120 TOPS

Q55. Accelerator dot product 1024 pairs of signed 8-bit integers. Minimum accumulator width:

- (a) 27 bits
- (b) 25 bits
- (c) 26 bits
- (d) 24 bits

Q56. Sensor $1.65 \text{ V} + 10 \text{ mV}/g \times a$, gain 100, 12-bit ADC ($0 - 3.3 \text{ V}$). Range and resolution:

- (a) $\pm 1.50 g$, 0.733 mg/LSB
- (b) $\pm 1.65 g$, 0.806 mg/LSB
- (c) $\pm 3.30 g$, 0.806 mg/LSB
- (d) $\pm 1.65 g$, 8.06 mg/LSB

Q57. 1500-byte packet across three 100 Mb/s links, 2.0 ms propagation per link. Total latency:

- (a) 6.36 ms
- (b) 6.48 ms

- (c) 6.12 ms
- (d) 6.24 ms

Q58. Host IP 192.168.10.77/27. Network, broadcast, usable range:

- (a) 192.168.10.64, 192.168.10.95, 192.168.10.65 – 94
- (b) 192.168.10.0, 192.168.10.127, 192.168.10.1 – 126
- (c) 192.168.10.64, 192.168.10.96, 192.168.10.65 – 95
- (d) 192.168.10.72, 192.168.10.87, 192.168.10.73 – 86

Q59. Fiber 80 km at 2.0×10^8 m/s vs Wireless 100 km at 3.0×10^8 m/s + 0.15 ms. Lower latency:

- (a) Fiber lower by 0.150 ms
- (b) Wireless lower by 0.167 ms
- (c) Wireless lower by 0.067 ms
- (d) Fiber lower by 0.083 ms

Q60. Workload 80% runs 20× on GPU, 20% on CPU, 5% transfer overhead. Overall speedup:

- (a) 3.20
- (b) 3.45
- (c) 4.00
- (d) 4.35

2.2 Section 2: Complete Answer Key & Technical Rationales

Part A: Questions 1 to 30

Table 2.1: Answer Key with Rationales (Questions 1 to 30)

Q#	Ans	Technical Rationale	Q#	Ans	Technical Rationale
1	C	$V = IR = 10 \text{ V}$.	16	B	Parallel matrix multiplication.
2	A	KCL states charge conservation at a node.	17	D	Converts physical parameter to electrical signal.
3	A	$12/2 = 6 \text{ V}$.	18	C	Connected computing nodes.
4	D	$V \propto R$.	19	D	Total internal reflection at core-cladding boundary.
5	B	$4/2 = 2 \text{ A}$.	20	C	GPU thousands of parallel cores.
6	D	Forward bias lowers potential barrier.	21	B	$12/6 = 2 \text{ A}$.
7	A	Forward bias reduces depletion width.	22	A	$5 + 2 = 4 + I \implies I = 3 \text{ A}$.
8	D	Converts AC to pulsating DC.	23	A	$20 \times 7/10 = 14 \text{ V}$.
9	A	Ideal forward diode has 0Ω resistance.	24	D	Parallel resistance drops, reducing voltage drop.
10	A	Emitter, Base, Collector.	25	B	$6 \times 6/9 = 4 \text{ A}$.
11	D	$I_C = \beta I_B$.	26	B	$V = 36 \text{ V} \implies I = 36/12 = 3 \text{ A}$.
12	C	Complementary NMOS and PMOS.	27	C	Depletion layer narrows.
13	A	Near-zero static standby power.	28	A	$V_{\text{applied}} = 0.6 \text{ V} < 0.7 \text{ V}$ threshold.
14	A	Volatile working memory.	29	B	$2 \times 50 = 100 \text{ Hz}$.
15	C	Non-volatile NAND flash.	30	D	Closed short circuit switch.

Part B: Questions 31 to 60

Table 2.2: Answer Key with Rationales (Questions 31 to 60)

Q#	Ans	Technical Rationale	Q#	Ans	Technical Rationale
31	C	$\beta = 2000/20 = 100$.	46	B	$V_{DC} = 13.46 \text{ V}$, $V_{r(pp)} = 0.269 \text{ V}$.
32	B	Both EBJ and CBJ forward biased.	47	C	Diode AND logic gate.
33	B	PMOS pulls output to V_{DD} .	48	D	$R_B = 2.5/0.009833 \approx 254 \Omega \Rightarrow 250 \Omega$.
34	C	One transistor always OFF in steady state.	49	B	$I_C = 99 \times \frac{1.30}{120} = 1.07 \text{ mA}$.
35	C	DRAM working memory.	50	B	$P_{orig} = 3.6 \text{ mW}$, $P_{new} = 2.0 \text{ mW} \Rightarrow 55.6\%$.
36	A	Solid-state silicon components.	51	B	A-pMOS pulls output HIGH.
37	B	Massively parallel SIMD processing.	52	A	$64 + 7 + 1 = 72 \text{ bits}$.
38	C	Conserves battery power.	53	D	$1.2 \times 3000/2.5 = 1.44 \text{ PB}$.
39	A	Subnet broadcast.	54	B	48 TOPS and 100 TOPS.
40	A	Total internal reflection in glass core.	55	A	25 – 27 bits signed accumulator.
41	B	KCL gives $V = 28/3 \text{ V}$, $I = 1/3 \text{ A}$.	56	B	$\pm 1.65 \text{ g}$ and 0.806 mg/LSB .
42	B	$R_p = 2 \text{ k}\Omega \Rightarrow R_L = 6 \text{ k}\Omega$.	57	A	$3(0.12) + 3(2.0) = 6.36 \text{ ms}$.
43	D	I_2 rises from 6.0 A to $7.2 \text{ A} \Rightarrow 1.2 \text{ A}$.	58	A	Subnet 64 – 95.
44	C	Combined I_S and thermal voltage exponential gives ≈ 2.79 .	59	D	Fiber 0.40 ms vs Wireless $0.483 \text{ ms} \Rightarrow 0.083 \text{ ms}$.
45	A	$I_D = 4.30/440 = 9.77 \text{ mA}$, $V_D = 0.798 \text{ V}$.	60	B	$1/(0.04 + 0.20 + 0.05) = 3.45$.

2.3 Section 3: Comprehensive Theory Questions & Worked Solutions

Theory Question 2.1

[University Exam / Analytical]

State KCL and KVL. For a loaded voltage divider (V_S, R_1, R_2, R_L), derive output voltage.

Comprehensive Step-by-Step Solution & Derivation:

KCL: sum of currents at a node is zero. KVL: sum of voltages around a loop is zero. Loaded divider output: $V_{out} = V_S \frac{R_2 \parallel R_L}{R_1 + R_2 \parallel R_L} = V_S \frac{R_2 R_L}{R_1(R_2 + R_L) + R_2 R_L}$.

Theory Question 2.2

[University Exam / Analytical]

Explain full-wave bridge rectifier with capacitor filter. Derive ripple voltage $V_{r(pp)} = \frac{I_{DC}}{2fC}$.

Comprehensive Step-by-Step Solution & Derivation:

Bridge rectifier conducts on both half-cycles ($f_r = 2f$). Capacitor charges to peak $V_m - 2V_D$ and discharges through load over $\Delta t = 1/(2f)$. Charge loss $\Delta Q = I_{DC}/(2f) = CV_{r(pp)} \Rightarrow V_{r(pp)} = \frac{I_{DC}}{2fC}$.

Chapter 3

Unit 3: Introduction to Number System and Logic Gates

3.1 Section 1: Chapter-Wise Multiple-Choice Questions (60 MCQs)

Part I: Foundational Level (Questions 1 to 20)

Q1. What is the decimal equivalent of binary 1010_2 ?

- (a) 14_{10}
- (b) 8_{10}
- (c) 12_{10}
- (d) 10_{10}

Q2. What is the binary equivalent of decimal 13_{10} ?

- (a) 1011_2
- (b) 1100_2
- (c) 1110_2
- (d) 1101_2

Q3. What is the Gray code equivalent of binary 1011_2 ?

- (a) 1010
- (b) 1110
- (c) 1001
- (d) 1101

Q4. What is the binary equivalent of Gray code 1101?

- (a) 1110
- (b) 1100
- (c) 1001
- (d) 1011

Q5. What is the Excess-3 code for decimal digit 4?

- (a) 0011
- (b) 1000
- (c) 0111
- (d) 0100

Q6. Which decimal digit is represented by Excess-3 code 1000?

- (a) 7
- (b) 4

- (c) 5
- (d) 6

Q7. Which BCD code represents decimal digit 9?

- (a) 1001
- (b) 1010
- (c) 1000
- (d) 1111

Q8. How is decimal number 25 represented in BCD?

- (a) 0010 0101
- (b) 0101 0010
- (c) 0010 0110
- (d) 0011 0101

Q9. What is the 1's complement of binary number 1010?

- (a) 1111
- (b) 0101
- (c) 0110
- (d) 1001

Q10. What is the 2's complement of the 4-bit binary number 0101?

- (a) 1110
- (b) 1100
- (c) 1011
- (d) 1010

Q11. What is the result of binary addition $101_2 + 011_2$?

- (a) 110_2
- (b) 100_2
- (c) 1000_2
- (d) 111_2

Q12. What is the result of binary subtraction $1101_2 - 0101_2$?

- (a) 1001_2
- (b) 1010_2
- (c) 1000_2
- (d) 0110_2

Q13. In 2's complement subtraction, what is added to the minuend?

- (a) The 2's complement of subtrahend
- (b) The 1's complement
- (c) The sign bit
- (d) The parity bit

Q14. Using 4-bit 2's complement, what is the result of $0110_2 - 0011_2$?

- (a) 1001_2
- (b) 1110_2
- (c) 0011_2
- (d) 0100_2

Q15. Which logic gate produces an output of 1 only when all inputs are 1?

- (a) AND gate

- (b) XOR gate
- (c) OR gate
- (d) NOT gate

Q16. What is the output of a NOT gate when its input is 0?

- (a) 0
- (b) 1
- (c) The same input
- (d) Undefined

Q17. According to Boolean identity law, what is $A + 0$?

- (a) A
- (b) 0
- (c) 1
- (d) $\bar{A}$

Q18. What does SOP stand for in Boolean algebra?

- (a) Sum of Products
- (b) Sequence of Products
- (c) Sum of Powers
- (d) Set of Parameters

Q19. Which expression is in Product of Sums (POS) form?

- (a) $A + B + C$
- (b) $AB + CD$
- (c) $ABC + DEF$
- (d) $(A + B)(C + D)$

Q20. What is the main purpose of a Karnaugh map?

- (a) To convert decimal numbers
- (b) To generate clock pulses
- (c) To simplify Boolean expressions
- (d) To store binary data

Part II: Intermediate Analytical Level (Questions 21 to 40)

Q21. What is the binary equivalent of decimal 173_{10} ?

- (a) 11001101_2
- (b) 10101101_2
- (c) 10101011_2
- (d) 10110101_2

Q22. Convert hexadecimal $3A7_{16}$ to decimal:

- (a) 947_{10}
- (b) 917_{10}
- (c) 935_{10}
- (d) 927_{10}

Q23. What is the binary equivalent of Gray code 1101?

- (a) 1010
- (b) 1111
- (c) 1001

(d) 1011

Q24. Convert binary 10110_2 into Gray code:

- (a) 11101
- (b) 10101
- (c) 11111
- (d) 10011

Q25. What is the Excess-3 representation of decimal 59?

- (a) 10011000
- (b) 01011001
- (c) 10001100
- (d) 01101001

Q26. The BCD code 1001 0110 represents which decimal number?

- (a) 86
- (b) 96
- (c) 150
- (d) 106

Q27. What is the 1's complement of 8-bit binary 01011010?

- (a) 10100101
- (b) 11011010
- (c) 10101001
- (d) 01000101

Q28. Result of binary addition $1011_2 + 1101_2$:

- (a) 10000_2
- (b) 11000_2
- (c) 10110_2
- (d) 11100_2

Q29. Using 5-bit 2's complement arithmetic, what is $10110_2 - 01001_2$?

- (a) 00101_2
- (b) 01101_2
- (c) 10011_2
- (d) 11101_2

Q30. Using 5-bit signed 2's complement, what is $01101_2 + 00111_2$?

- (a) 01010_2
- (b) 10100_2
- (c) 11100_2
- (d) 10010_2

Q31. A NAND gate has inputs $A = 1$ and $B = 1$. Output is:

- (a) 1
- (b) 0
- (c) A
- (d) B

Q32. Which gate produces output 1 only when two inputs are different?

- (a) XOR gate
- (b) OR gate

- (c) AND gate
- (d) XNOR gate

Q33. Simplify Boolean expression $A + AB$:

- (a) $A + B$
- (b) AB
- (c) A
- (d) B

Q34. Using De Morgan's theorem, simplify $(A + B)'$:

- (a) AB
- (b) $(AB)'$
- (c) $A'B'$
- (d) $A' + B'$

Q35. Which of the following is in standard Sum of Products (SOP) form?

- (a) $(A + B') + C$
- (b) $A(B + C')$
- (c) $A'B + AC + BC'$
- (d) $(A + B)(C + D)$

Q36. Which is in Product of Sums (POS) form?

- (a) $(A + B')(A' + C)$
- (b) $AB(C + D)$
- (c) $A(B + C)$
- (d) $AB + A'C$

Q37. Using 3-variable K-map, simplify $F(A, B, C) = \sum m(1, 3, 5, 7)$:

- (a) A
- (b) $A + B$
- (c) C
- (d) B

Q38. Using 4-variable K-map for $F = \sum m(0, 2, 8, 10)$:

- (a) $B'C'$
- (b) $A'D'$
- (c) BD'
- (d) $B'D'$

Q39. Using 4-variable K-map for $F = \sum m(1, 3, 9, 11)$:

- (a) $A'D$
- (b) $B'C'$
- (c) BC'
- (d) $B'D$

Q40. Simplify Boolean expression $(A + B)(A + C)$:

- (a) $A + BC$
- (b) ABC
- (c) $A + B + C$
- (d) $AB + AC$

Part III: Advanced Mastery Level (Questions 41 to 60)

Q41. Convert hexadecimal $(3A7.C)_{16}$ to octal:

- (a) $(1647.3)_8$
- (b) $(1647.6)_8$
- (c) $(1657.4)_8$
- (d) $(1727.6)_8$

Q42. Which base-7 representation is exactly equivalent to $(231.2)_4$?

- (a) $(63.\bar{4})_7$
- (b) $(64.3)_7$
- (c) $(63.\bar{3})_7$
- (d) $(63.3)_7$

Q43. 6-bit Gray 101101 received with 3rd bit from left inverted. Decimal value represented by corrupted Gray word:

- (a) 59
- (b) 55
- (c) 53
- (d) 57

Q44. What is binary equivalent of reflected Gray 11010111?

- (a) 11001010
- (b) 10010110
- (c) 10110010
- (d) 10011010

Q45. Two Excess-3 numbers 1001 0100 and 0111 1010. Excess-3 representation of their decimal sum:

- (a) 0100 0011 1011
- (b) 0100 0011 1000
- (c) 0001 0000 1000
- (d) 0100 0110 1011

Q46. Two BCD digits 1000 and 0111 added with carry-in 1. Output carry and BCD sum digit after correction:

- (a) Carry 0, digit 0110
- (b) Carry 1, digit 0110
- (c) Carry 0, digit 1001
- (d) Carry 1, digit 0000

Q47. Using 8-bit one's complement, what bit pattern results from $37-92$ before interpreting sign?

- (a) 11001001
- (b) 11001000
- (c) 10100111
- (d) 00110111

Q48. Fixed-point octal register (6 integer, 2 fraction). 8's complement of $(000527.40)_8$:

- (a) $(777250.40)_8$
- (b) $(777470.40)_8$
- (c) $(777251.40)_8$

(d) $(777250.37)_8$

Q49. Evaluate unsigned binary product $(110101)_2 \times (1011)_2$:

- (a) $(1000111111)_2$
- (b) $(1010000111)_2$
- (c) $(1001000111)_2$
- (d) $(1001010111)_2$

Q50. Unsigned binary $(101101101)_2$ divided by $(1101)_2$. Quotient and remainder:

- (a) Quotient 11101, rem 0
- (b) Quotient 11100, rem 11
- (c) Quotient 11011, rem 10
- (d) Quotient 11100, rem 1

Q51. 8-bit two's complement arithmetic for $45 - (-83)$:

- (a) 01111111, overflow
- (b) 10000000, overflow
- (c) 00000000, no overflow
- (d) 10000000, no overflow

Q52. 6-bit two's complement processor evaluates $(-27) - (+14)$:

- (a) 101001, overflow
- (b) 010111, no overflow
- (c) 100111, no overflow
- (d) 010111, overflow

Q53. NAND logic: $X = A \text{ NAND } B, Y = A \text{ NAND } X, Z = B \text{ NAND } X, F = Y \text{ NAND } Z$. Function F :

- (a) AB
- (b) $A \oplus B$
- (c) $A \odot B$
- (d) $A + B$

Q54. NOR logic: $X = A \text{ NOR } B, Y = A \text{ NOR } X, Z = B \text{ NOR } X, F = Y \text{ NOR } Z$. Function F :

- (a) $A \oplus B$
- (b) $A \odot B$
- (c) $A + B$
- (d) $(AB)'$

Q55. Simplify $F = (A + B)(A' + C)(B + C)$ using Boolean algebra:

- (a) $(A + B)(A' + C)$
- (b) $(A + C)(A' + B)$
- (c) $(A + B)(B + C)$
- (d) $(A' + C)(B + C)$

Q56. Which expression is equivalent to $F = A'B + AC + BC$?

- (a) $AC + BC$
- (b) $A'B + BC$
- (c) $A'B + AC$
- (d) $A'B + A'C$

Q57. Canonical POS representation of $F(A, B, C, D) = \sum m(0, 2, 5, 7, 8, 10, 13, 15)$:

- (a) $\prod M(1, 3, 4, 6, 9, 11, 12, 14)$

- (b) $\prod M(0, 2, 5, 7, 8, 10, 13, 15)$
- (c) $\prod M(1, 3, 5, 7, 9, 11, 13, 15)$
- (d) $\prod M(2, 4, 6, 8, 10, 12, 14, 15)$

Q58. Convert $F = (A + B')(B + C)$ to canonical SOP form for A, B, C :

- (a) $\sum m(0, 2, 3, 4)$
- (b) $\sum m(1, 3, 5, 7)$
- (c) $\sum m(1, 5, 6, 7)$
- (d) $\sum m(2, 4, 6, 7)$

Q59. 4-variable K-map: $F = \sum m(1, 3, 9, 11, 12, 14) + d(0, 2, 8, 10)$ in SOP form:

- (a) $B' + AD$
- (b) $B' + A'D'$
- (c) $B'D + AD'$
- (d) $B' + AD'$

Q60. Minimal 2-level SOP $F = \sum m(4, 5, 6, 7, 10, 11, 14, 15) = A'B + AC$ has static-1 hazard. Hazard-free expression:

- (a) $A'B + AC + BC$
- (b) $A'B + AC + A'C$
- (c) $A'B + AC + B'C$
- (d) $A'B + AC + AB$

3.2 Section 2: Complete Answer Key & Technical Rationales

Part A: Questions 1 to 30

Table 3.1: Answer Key with Rationales (Questions 1 to 30)

Q#	Ans	Technical Rationale	Q#	Ans	Technical Rationale
1	D	$1010_2 = 1 \times 8 + 0 \times 4 + 1 \times 2 + 0 = 10_{10}$.	16	B	NOT inverts logic level: $\bar{0} = 1$.
2	D	$13 = 8 + 4 + 1 = 1101_2$.	17	A	$A + 0 = A$.
3	B	$G_3 = 1, G_2 = 1 \oplus 0 = 1, G_1 = 0 \oplus 1 = 1, G_0 = 1 \oplus 1 = 0 \implies 1110$.	18	A	SOP = Sum of Products (OR of AND terms).
4	C	$B_3 = 1, B_2 = 1 \oplus 1 = 0, B_1 = 0 \oplus 0 = 0, B_0 = 0 \oplus 1 = 1 \implies 1001_2$.	19	D	POS is an AND of OR clauses.
5	C	$4 + 3 = 7 = 0111_2$.	20	C	K-Maps graphically minimize Boolean switching functions.
6	C	$1000_2 = 8; 8 - 3 = 5$.	21	B	$173 = 128 + 32 + 8 + 4 + 1 = 10101101_2$.
7	A	8421 code for 9 is 1001_2 .	22	C	$3 \times 256 + 10 \times 16 + 7 = 768 + 160 + 7 = 935_{10}$.
8	A	$2 = 0010_2, 5 = 0101_2$.	23	C	$B_3 = 1, B_2 = 1 \oplus 1 = 0, B_1 = 0 \oplus 0 = 0, B_0 = 0 \oplus 1 = 1 \implies 1001_2$.
9	B	Invert all bits: $1010 \rightarrow 0101$.	24	A	$G_4 = 1, G_3 = 1 \oplus 0 = 1, G_2 = 0 \oplus 1 = 1, G_1 = 1 \oplus 1 = 0, G_0 = 1 \oplus 0 = 1 \implies 11101$.
10	C	1's comp = 1010 ; $1010 + 1 = 1011$.	25	C	$5 + 3 = 8 = 1000_2, 9 + 3 = 12 = 1100_2 \implies 10001100$.
11	C	$5 + 3 = 8 = 1000_2$.	26	B	$1001_2 = 9, 0110_2 = 6 \implies 96$.
12	C	$13 - 5 = 8 = 1000_2$.	27	A	Bitwise NOT: $01011010 \rightarrow 10100101$.
13	A	$A - B = A + (2\text{'s comp of } B)$.	28	B	$11 + 13 = 24 = 11000_2$.
14	C	$6 - 3 = 3 = 0011_2$.	29	B	$22 - 9 = 13 = 01101_2$.
15	A	Standard definition of AND gate.	30	B	$+13 + (+7) = +20$. In 5 bits, range is $[-16, +15]$; $+20 = 10100_2$ (overflow occurs, stored pattern 10100).

Part B: Questions 31 to 60

Table 3.2: Answer Key with Rationales (Questions 31 to 60)

Q#	Ans	Technical Rationale	Q#	Ans	Technical Rationale
31	B	$\overline{1} \cdot \overline{1} = 0$.	46	B	$8 + 7 + 1 = 16_{10}$. Binary sum = 10000_2 . Add $0110_2 \Rightarrow 10110_2$ (Carry 1, digit 6).
32	A	XOR is the difference/inequality detector ($A \neq B$).	47	A	$37 = 00100101_2$; $92 = 01011100_2$. 1's comp of 92 is 10100011_2 . Sum = $00100101 + 10100011 = 11001000_2$.
33	C	$A(1+B) = A(1) = A$ (Absorption law).	48	A	7's complement = $(777250.37)_8$. Add 1 at LSB (0.01) $\Rightarrow (777250.40)_8$.
34	C	$\overline{A+B} = \bar{A}\bar{B}$.	49	C	$53 \times 11 = 583 = 1001000111_2$.
35	C	Sum (OR) of product (AND) terms.	50	D	$365/13 = 28 \text{ rem } 1$. $28 = 11100_2$, rem 1_2 .
36	A	Product (AND) of sum (OR) terms.	51	B	$45 - (-83) = +128$. 8-bit max is $+127$. Result wraps to 10000000_2 with signed overflow flag set.
37	C	Minterms 1, 3, 5, 7 correspond to $C = 1$ independent of A, B .	52	D	$-27 - 14 = -41$. 6-bit range is $[-32, +31]$. Signed underflow occurs; stored bits 010111_2 with overflow flag.
38	D	Corner cells 0, 2, 8, 10 form a quad = $\bar{B}\bar{D}$.	53	B	This is the classic 4-NAND implementation of XOR ($A \oplus B$).
39	D	Cells in columns 01, 11 for rows 00, 10 form a quad = $\bar{B}D$.	54	B	Dual 4-NOR circuit implements XNOR ($A \odot B$).
40	A	$(A+B)(A+C) = AA + AC + AB + BC = A(1+C+B) + BC = A + BC$ (Distributive law).	55	A	By consensus theorem in POS form, $(B+C)$ is redundant consensus term.
41	B	$0011\ 1010\ 0111.1100_2 \rightarrow (1647.6)_8$.	56	C	By consensus theorem, BC is redundant consensus term.
42	A	$(231.2)_4 = 2(16) + 3(4) + 1 + 2/4 = 45.5_{10}$. $45_{10} = (63)_7$; $0.5_{10} = (0.\bar{3}\bar{3})_7$ or $(63.\bar{3})_7$.	57	A	Maxterms are the complement set of minterm indices: $\{1, 3, 4, 6, 9, 11, 12, 14\}$.
43	B	Corrupted: 100101. Binary: $B_5 = 1, B_4 = 1, B_3 = 1, B_2 = 0, B_1 = 0, B_0 = 1 \Rightarrow 111001_2 = 57_{10}$ (or 55 depending on index).	58	C	$(A + \bar{B})(B + C) = AB + AC + \bar{B}C = \sum m(1, 5, 6, 7)$.
44	D	XOR sequentially: 10011010_2 .	59	D	Octet covering rows 00, 10 gives $\bar{B}$; quad covering 12, 14, 8, 10 gives $A\bar{D}$.
45	A	Decimals: $(9-3)(4-3) = 61$; $(7-3)(10-3) = 47$. Sum = $61 + 47 = 108$. XS-3 of 1, 0, 8 is 0100 0011 1011.	60	A	Adding the consensus term BC bridges adjacent groups and eliminates the static-1 hazard.

3.3 Section 3: Comprehensive Theory Questions & Worked Solutions

Theory Question 3.1

[University Exam / Analytical]

State and prove De Morgan's Laws of Boolean Algebra using truth tables.

Comprehensive Step-by-Step Solution & Derivation:

1. **First Law:** $\overline{A + B} = \bar{A} \cdot \bar{B}$.

2. **Second Law:** $\overline{A \cdot B} = \bar{A} + \bar{B}$.

Proof by perfect induction (truth table verification for all 4 states shows identical columns for LHS and RHS).

Theory Question 3.2

[University Exam / Analytical]

Explain the concept of Static-1 Hazards in combinational circuits and show how to design hazard-free logic using Karnaugh Maps.

Comprehensive Step-by-Step Solution & Derivation:

A static-1 hazard is a transient 0 glitch that occurs during input transitions between adjacent 1s located in different prime implicant groups due to unequal gate propagation delays. It is eliminated by including the redundant consensus prime implicant covering the boundary cells.

Chapter 4

Unit 4: Introduction to Combinational Logic Circuits

4.1 Section 1: Chapter-Wise Multiple-Choice Questions (60 MCQs)

Part I: Foundational Level (Questions 1 to 20)

Q1. What are the two outputs of a half adder?

- (a) Input and output
- (b) Select and enable
- (c) Difference and borrow
- (d) Sum and carry

Q2. Which logic operation produces the sum output of a half adder?

- (a) OR operation
- (b) NOT operation
- (c) AND operation
- (d) XOR operation

Q3. How many input bits does a full adder accept?

- (a) Three input bits
- (b) One input bit
- (c) Two input bits
- (d) Four input bits

Q4. What are the two outputs of a half subtractor?

- (a) Difference and borrow
- (b) Sum and carry
- (c) Quotient and remainder
- (d) Product and carry

Q5. Which logic operation produces the difference output of a half subtractor?

- (a) XOR operation
- (b) AND operation
- (c) NOT operation
- (d) NOR operation

Q6. What additional input distinguishes a full subtractor from a half subtractor?

- (a) Borrow-in input
- (b) Carry-in input

- (c) Select input
- (d) Enable input

Q7. What is the main function of a multiplexer?

- (a) Select one input
- (b) Store one input
- (c) Activate every input
- (d) Invert every input

Q8. How many select lines are required by a 4-to-1 multiplexer?

- (a) Four select lines
- (b) Two select lines
- (c) One select line
- (d) Three select lines

Q9. How many outputs does a basic multiplexer have?

- (a) One output
- (b) Two outputs
- (c) Eight outputs
- (d) Four outputs

Q10. What is the main function of a de-multiplexer?

- (a) Combine many inputs
- (b) Store many inputs
- (c) Compare two inputs
- (d) Route one input to many outputs

Q11. How many data inputs does a basic de-multiplexer have?

- (a) Two data inputs
- (b) One data input
- (c) Four data inputs
- (d) Eight data inputs

Q12. How many output lines are present in a 1-to-4 de-multiplexer?

- (a) Two output lines
- (b) Four output lines
- (c) Three output lines
- (d) Eight output lines

Q13. How many outputs does a 2-to-4 decoder have?

- (a) Three outputs
- (b) Four outputs
- (c) Two outputs
- (d) Eight outputs

Q14. For each valid input combination, how many outputs of a standard binary decoder are active?

- (a) One output
- (b) No outputs
- (c) Two outputs
- (d) All outputs

Q15. What is the purpose of an enable input on a decoder?

- (a) Control decoder operation
- (b) Increase input count
- (c) Store decoded data
- (d) Invert output values

Q16. What does a binary encoder produce from an active input line?

- (a) A stored bit
- (b) A clock pulse
- (c) An analog signal
- (d) A binary code

Q17. How many output bits does an 8-to-3 encoder produce?

- (a) Three output bits
- (b) Two output bits
- (c) Eight output bits
- (d) Four output bits

Q18. What is the purpose of a digital comparator?

- (a) Store binary numbers
- (b) Add binary numbers
- (c) Compare binary numbers
- (d) Encode binary numbers

Q19. Which output is active when two comparator inputs have the same value?

- (a) $A > B$
- (b) $A \neq B$
- (c) $A < B$
- (d) $A = B$

Q20. For 2-bit numbers $A = 10_2$ and $B = 01_2$, which comparison is correct?

- (a) $A = 0$
- (b) $A > B$
- (c) $A < B$
- (d) $A = B$

Part II: Intermediate Analytical Level (Questions 21 to 40)

Q21. Half adder with $A = 1, B = 1$. Outputs S, C are:

- (a) $S = 0, C = 1$
- (b) $S = 1, C = 1$
- (c) $S = 1, C = 0$
- (d) $S = 0, C = 0$

Q22. Full adder with $A = 1, B = 0, C_{in} = 1$. Outputs are:

- (a) $S = 0, C_{out} = 1$
- (b) $S = 1, C_{out} = 1$
- (c) $S = 0, C_{out} = 0$
- (d) $S = 1, C_{out} = 0$

Q23. 4-bit ripple carry adder adds 0111_2 and 1001_2 with $C_{in} = 1$. Result:

- (a) $C_{out} = 0, S = 0001$
- (b) $C_{out} = 1, S = 0000$

- (c) $C_{out} = 0, S = 1111$
 (d) $C_{out} = 1, S = 0001$

Q24. Full subtractor receives $A = 0, B = 1, B_{in} = 1$. Difference and borrow-out:

- (a) $D = 1, B_{out} = 0$
 (b) $D = 0, B_{out} = 1$
 (c) $D = 1, B_{out} = 1$
 (d) $D = 0, B_{out} = 0$

Q25. 2-bit subtractor calculates $10_2 - 01_2$. Difference and final borrow-out:

- (a) $D = 01, B_{out} = 0$
 (b) $D = 11, B_{out} = 0$
 (c) $D = 01, B_{out} = 1$
 (d) $D = 00, B_{out} = 1$

Q26. Which pair of Boolean expressions correctly represents difference and borrow of half subtractor?

- (a) $D = A \oplus B, B_{out} = \bar{A}B$
 (b) $D = A \oplus B, B_{out} = A\bar{B}$
 (c) $D = \bar{A} \oplus B, B_{out} = AB$
 (d) $D = AB, B_{out} = \bar{A}B$

Q27. 4-to-1 MUX has $I_0 = 0, I_1 = 1, I_2 = 1, I_3 = 0$. Output when $S_1S_0 = 10$:

- (a) 1, determined by I_1
 (b) 0, determined by I_3
 (c) 0, determined by I_0
 (d) 1, determined by I_2

Q28. 2-to-1 MUX uses B as select input. Which input assignments make output $A \oplus B$?

- (a) $I_0 = A, I_1 = A$
 (b) $I_0 = \bar{A}, I_1 = \bar{A}$
 (c) $I_0 = A, I_1 = \bar{A}$
 (d) $I_0 = \bar{A}, I_1 = A$

Q29. Function $F(A, B, C) = \sum m(1, 2, 5, 6)$ with MUX selects A, B . Inputs (I_0, I_1, I_2, I_3) :

- (a) $C, C, \bar{C}, \bar{C}$
 (b) $C, \bar{C}, C, \bar{C}$
 (c) $\bar{C}, \bar{C}, C, C$
 (d) $\bar{C}, C, \bar{C}, C$

Q30. 1-to-4 DEMUX with data $D = 1$ and select $S_1S_0 = 11$. Active output:

- (a) $Y_1 = 1$
 (b) $Y_3 = 1$
 (c) $Y_0 = 1$
 (d) $Y_2 = 1$

Q31. How to use 1-to-8 DEMUX as 3-to-8 active-high decoder?

- (a) Connect outputs together
 (b) Fix data input at 1
 (c) Use select as data
 (d) Fix data input at 0

Q32. 3-to-8 active-high decoder receives 110_2 . Which output is active?

- (a) Y_6
- (b) Y_5
- (c) Y_7
- (d) Y_3

Q33. 2-to-4 decoder implements $F(A, B) = \sum m(1, 3)$. Which outputs must be ORed?

- (a) Y_1 and Y_3
- (b) Y_0 and Y_1
- (c) Y_0 and Y_2
- (d) Y_2 and Y_3

Q34. How many 2-to-4 decoders with enables construct one 3-to-8 decoder?

- (a) Four decoders
- (b) One decoder
- (c) Three decoders
- (d) Two decoders

Q35. In 8-to-3 binary encoder, only input D_6 is active. Encoded output $Y_2Y_1Y_0$:

- (a) 011
- (b) 111
- (c) 110
- (d) 101

Q36. 8-to-3 priority encoder (D_7 highest). If $D_5 = 1$ and $D_2 = 1$ with others 0, output is:

- (a) 111
- (b) 000
- (c) 010
- (d) 101

Q37. 2-bit comparator with $A = 10_2, B = 01_2$. Active output:

- (a) $A = B$
- (b) None
- (c) $A > B$
- (d) $A < B$

Q38. 2-bit comparator reports $A > B = 0, A = B = 0, A < B = 1$. If $A = 10_2$, value of B could be:

- (a) 00_2
- (b) 10_2
- (c) 11_2
- (d) 01_2

Q39. For $A = A_1A_0$ and $B = B_1B_0$, correct expression for $A > B$:

- (a) $A_1\bar{B}_1 + (A_1 \odot B_1)A_0\bar{B}_0$
- (b) $\bar{A}_1B_1 + (A_1 \odot B_1)\bar{A}_0B_0$
- (c) $A_1\bar{B}_1 + (A_1 \oplus B_1)\bar{A}_0B_0$
- (d) $A_1\bar{B}_1 + (A_1 \oplus B_1)A_0B_0$

Q40. Two 2-bit numbers have equal MSBs. If $A_0 = 0$ and $B_0 = 1$, comparator indicates:

- (a) $A < B$
- (b) $A > B$
- (c) Undefined

(d) $A = B$

Part III: Advanced Mastery Level (Questions 41 to 60)

Q41. 4-bit ripple adder adds 0111 and 0001 with $C_0 = 0$. Each carry delay is t_c , sum delay t_s . Guaranteed valid time:

- (a) $4t_c + t_s$
- (b) $\max(2t_c + t_s, 3t_c)$
- (c) $\max(3t_c + t_s, 4t_c)$
- (d) $3t_c + 4t_s$

Q42. For 2-bit CLA, carry C_2 without intermediate C_1 :

- (a) $C_2 = G_1P_1 + G_0P_0 + P_1C_0$
- (b) $C_2 = G_1 + P_0G_0 + P_1P_0C_0$
- (c) $C_2 = G_0 + P_0G_1 + P_1P_0C_0$
- (d) $C_2 = G_1 + P_1G_0 + P_1P_0C_0$

Q43. 4-bit two's complement adder computes $0110 + 0101$ (+6 + +5). Sum, unsigned carry-out, signed overflow flag:

- (a) $S = 1011, C_{out} = 0, V = 1$
- (b) $S = 1011, C_{out} = 1, V = 0$
- (c) $S = 1011, C_{out} = 0, V = 0$
- (d) $S = 0011, C_{out} = 1, V = 1$

Q44. Full subtractor borrow-out using $A \oplus B$ propagation term:

- (a) $B_{out} = \bar{A}B + (A \oplus B)B_{in}$
- (b) $B_{out} = A\bar{B} + \overline{A \oplus B}B_{in}$
- (c) $B_{out} = \bar{A}B + \overline{A \oplus B}B_{in}$
- (d) $B_{out} = A\bar{B} + (A \oplus B)B_{in}$

Q45. 4-bit adder performs $A - B$ as $A + \bar{B} + 1$ with $A = 0011, B = 0101$. Result D, C_{out}, B_{out}, V :

- (a) $D = 1110, C_{out} = 0, B_{out} = 1, V = 0$
- (b) $D = 1110, C_{out} = 1, B_{out} = 0, V = 0$
- (c) $D = 0010, C_{out} = 1, B_{out} = 0, V = 1$
- (d) $D = 1110, C_{out} = 0, B_{out} = 1, V = 1$

Q46. 4-bit two's complement subtractor computes $A - B$ for $A = 1000(-8)$ and $B = 0001(+1)$:

- (a) $D = 1001, B_{out} = 1, V = 0$
- (b) $D = 1111, B_{out} = 0, V = 1$
- (c) $D = 0111, B_{out} = 1, V = 0$
- (d) $D = 0111, B_{out} = 0, V = 1$

Q47. Function $F(A, B, C) = \sum m(1, 2, 5, 7)$ with MUX selects B, C (00, 01, 10, 11). Inputs (I_0, I_1, I_2, I_3) :

- (a) $(0, 1, \bar{A}, A)$
- (b) $(\bar{A}, A, 0, 1)$
- (c) $(0, 1, A, \bar{A})$
- (d) $(A, \bar{A}, 1, 0)$

Q48. 4-to-1 MUX selects A, B to implement majority $M = AB + AC + BC$. For $AB = 00, 01, 10, 11$, data inputs:

- (a) $(C, C, 0, 1)$

- (b) $(0, C, C, 1)$
- (c) $(0, \bar{C}, C, 1)$
- (d) $(C, 0, 1, C)$

Q49. 16-to-1 MUX using 4-to-1 MUXes without external gates. Minimum multiplexers and levels:

- (a) 4 in 2 levels
- (b) 5 in 2 levels
- (c) 6 in 2 levels
- (d) 5 in 3 levels

Q50. Active-high 1-to-8 DEMUX built from one 1-to-2 and two 1-to-4 DEMUXes. Preserving select ordering $S_2S_1S_0$:

- (a) Use S_1 on 1:2 and S_2S_0 on 1:4
- (b) Use S_0 on 1:2 and S_2S_1 on 1:4
- (c) Use S_2 on 1:2 and S_1S_0 on 1:4
- (d) Use S_2S_1 on 1:4 and connect S_0

Q51. 1-to-8 DEMUX with inputs ABC , outputs Y_1, Y_2, Y_7 ORed. Simplified function:

- (a) $D[A(B \oplus C) + \bar{A}BC]$
- (b) $D[\bar{A}(B \odot C) + ABC]$
- (c) $D[A(B \odot C) + \bar{A}BC]$
- (d) $D[\bar{A}(B \oplus C) + ABC]$

Q52. 3-to-8 decoder with active-high outputs implements $F(A, B, C) = \prod M(0, 3, 5, 6)$. Gate connection:

- (a) $F = Y_1 + Y_2 + Y_4 + Y_7$
- (b) $F = Y_0Y_3Y_5Y_6$
- (c) $F = \overline{Y_0 + Y_3 + Y_5 + Y_6}$
- (d) $F = Y_0 + Y_3 + Y_5 + Y_6$

Q53. Two 2-to-4 decoders with active-low enable $\bar{E}$ to make 3-to-8 decoder with MSB A :

- (a) Lower $\bar{E} = A$, upper $\bar{E} = \bar{A}$
- (b) Both connected to A
- (c) Both connected to $\bar{A}$
- (d) Lower $\bar{E} = \bar{A}$, upper $\bar{E} = A$

Q54. 3-to-8 decoder changes input directly from 011 to 100. What can occur due to unequal delays?

- (a) Transient interval with no active output or multiple active outputs
- (b) Must move directly with no transient
- (c) Only Y_3 glitches
- (d) Only Y_4 glitches

Q55. Non-priority 4-to-2 encoder: $Q_1 = D_2 + D_3, Q_0 = D_1 + D_3$. If $D_1 = D_2 = 1$ with others 0, output is:

- (a) $Q_1Q_0 = 11$, falsely resembling active D_3
- (b) $Q_1Q_0 = 10$
- (c) $Q_1Q_0 = 01$
- (d) $Q_1Q_0 = 00$

Q56. 4-to-2 priority encoder ($D_3 > D_2 > D_1 > D_0$). Correct equations for Q_1Q_0 and valid V :

- (a) $Q_1 = D_3 + D_2, Q_0 = D_3 + \bar{D}_2 D_1, V = \sum D_i$
- (b) $Q_1 = D_3 + \bar{D}_2, Q_0 = D_1 + D_0, V = \sum D_i$
- (c) $Q_1 = D_3 + D_2, Q_0 = D_3 + D_1, V = \prod D_i$
- (d) $Q_1 = D_2 + D_1, Q_0 = D_3 + \bar{D}_2 D_0, V = \sum D_i$

Q57. Two 8-to-3 priority encoders cascaded to 16-to-4 (D_{15} highest). High-level combination:

- (a) $Q_3 = V_H$, select high code when $V_H = 1, V = V_H + V_L$
- (b) $Q_3 = V_H V_L$, select low code when $V_L = 0$
- (c) $Q_3 = V_L$, select low code
- (d) $Q_3 = V_H + V_L$, always high

Q58. For unsigned 2-bit numbers A, B , Boolean expression for $A = B$:

- (a) $(A_1 \odot B_1)(A_0 \odot B_0)$
- (b) $(A_1 \oplus B_1)(A_0 \oplus B_0)$
- (c) $(A_1 \odot B_1) + (A_0 \odot B_0)$
- (d) $(A_1 \oplus B_1) + (A_0 \oplus B_0)$

Q59. Unsigned condition $A > B$ for 2-bit numbers:

- (a) $A_1 \bar{B}_1 + (A_1 \oplus B_1) A_0 \bar{B}_0$
- (b) $A_1 \bar{B}_1 + (A_1 \odot B_1) A_0 \bar{B}_0$
- (c) $A_0 \bar{B}_0 + (A_0 \odot B_0) A_1 \bar{B}_1$
- (d) $A_1 \bar{B}_1 + (A_1 \odot B_1) \bar{A}_0 B_0$

Q60. Inputs $A = 10$ and $B = 01$ applied to 2-bit comparator. Unsigned vs 2's complement results:

- (a) Unsigned gives $A > B$; signed gives $A < B$
- (b) Unsigned gives $A > B$; signed gives $A = B$
- (c) Unsigned gives $A < B$; signed gives $A > B$
- (d) Unsigned gives $A = B$; signed gives $A < B$

4.2 Section 2: Complete Answer Key & Technical Rationales

Part A: Questions 1 to 30

Table 4.1: Answer Key with Rationales (Questions 1 to 30)

Q#	Ans	Technical Rationale	Q#	Ans	Technical Rationale
1	D	A half adder generates Sum ($S = A \oplus B$) and Carry ($C = AB$).	16	D	Encodes 2^n inputs into an n -bit binary address.
2	D	$S = A \oplus B$.	17	A	$8 = 2^3 \implies 3$ output bits.
3	A	Full adder takes inputs A, B , and C_{in} .	18	C	Determines whether $A > B, A = B$, or $A < B$.
4	A	Outputs are Difference ($D = A \oplus B$) and Borrow ($B_{out} = \bar{A}B$).	19	D	Equality line $A = B$ is asserted.
5	A	$D = A \oplus B$.	20	B	$A = 2, B = 1 \implies A > B$.
6	A	Full subtractor accepts Borrow-in (B_{in}).	21	A	$1 + 1 = 10_2 \implies S = 0, C = 1$.
7	A	A multiplexer routes one of many data inputs to a single output line.	22	A	$1 + 0 + 1 = 2_{10} = 10_2 \implies S = 0, C_{out} = 1$.
8	B	$2^n = 4 \implies n = 2$ select lines.	23	D	$7 + 9 + 1 = 17_{10} = 10001_2 \implies C_{out} = 1, S = 0001_2$.
9	A	A multiplexer has multiple inputs and exactly one output.	24	B	$0 - 1 - 1 = -2_{10} \implies D = 0, B_{out} = 1$.
10	D	Routes a single input data line to one of 2^n output lines.	25	A	$2 - 1 = 1 = 01_2$ with no borrow ($B_{out} = 0$).
11	B	A demux has a single data input.	26	A	$D = A \oplus B, B_{out} = \bar{A}B$.
12	B	1-to-4 demux has 4 output lines.	27	D	Select $10_2 = 2$ routes input $I_2 = 1$ to output.
13	B	$2^2 = 4$ outputs.	28	C	When $B = 0, Y = A$; when $B = 1, Y = \bar{A} \implies Y = \bar{B}A + B\bar{A} = A \oplus B$.
14	A	Decoders are one-hot: exactly one output line is active per valid input.	29	B	For $AB = 00, Y = C$; for $AB = 01, Y = \bar{C}$; for $AB = 10, Y = C$; for $AB = 11, Y = \bar{C}$.
15	A	Enable line activates or disables the entire chip.	30	B	$11_2 = 3 \implies Y_3 = D = 1$.

Part B: Questions 31 to 60

Table 4.2: Answer Key with Rationales (Questions 31 to 60)

Q#	Ans	Technical Rationale	Q#	Ans	Technical Rationale
31	B	With $D = 1$, select lines decode directly to active-high outputs $Y_k = m_k$.	46	D	$-8 - (+1) = -9 < -8$ (underflow). Result stored = $0111_2(+7)$ with $V = 1$.
32	A	$110_2 = 6 \implies Y_6 = 1$.	47	A	$BC = 00 \rightarrow 0; BC = 01 \rightarrow 1; BC = 10 \rightarrow \bar{A}; BC = 11 \rightarrow A$.
33	A	$F = Y_1 + Y_3 = m_1 + m_3$.	48	B	$AB = 00 \rightarrow 0; AB = 01 \rightarrow C; AB = 10 \rightarrow C; AB = 11 \rightarrow 1$.
34	D	Two 2-to-4 decoders enabled by A and $\bar{A}$.	49	B	4 first-level 4:1 MUXes fed into 1 second-level 4:1 MUX = 5 total in 2 levels.
35	C	$6_{10} = 110_2$.	50	C	MSB S_2 selects which 1-to-4 bank is active; S_1S_0 select the individual output line.
36	D	Highest active index is 5 $\implies 101_2$.	51	D	$Y_1 + Y_2 + Y_7 = D(\bar{A}\bar{B}C + \bar{A}B\bar{C} + ABC) = D[\bar{A}(B \oplus C) + ABC]$.
37	C	$2 > 1 \implies A > B$ active.	52	A	Complement of maxterms 0, 3, 5, 6 are minterms 1, 2, 4, 7 $\implies F = Y_1 + Y_2 + Y_4 + Y_7$.
38	C	$A < B \implies B > 2 \implies B = 3 = 11_2$.	53	D	For lower half ($A = 0$), enable must be 0 $\implies \bar{E}_{lower} = A$. For upper half ($A = 1$), $\bar{E}_{upper} = \bar{A}$.
39	A	MSB check $A_1\bar{B}_1$ or if MSBs match ($A_1 \odot B_1$), LSB check $A_0\bar{B}_0$.	54	A	Because multiple bits change simultaneously ($011 \rightarrow 100$), intermediate transient addresses produce spurious glitches.
40	A	With MSBs equal, $A_0 < B_0 \implies A < B$.	55	A	$Q_1 = 1 + 0 = 1, Q_0 = 1 + 0 = 1 \implies 11_2$, which falsely reports input 3.
41	C	Carry ripples to stage 3 at $3t_c$, generating S_3 at $3t_c + t_s$, while C_4 completes at $4t_c$.	56	A	Standard priority encoder synthesis equations.
42	D	Standard CLA formula: $C_2 = G_1 + P_1(G_0 + P_0C_0) = G_1 + P_1G_0 + P_1P_0C_0$.	57	A	If high group has valid active input ($V_H = 1$), $Q_3 = 1$ and upper encoder outputs are selected; overall valid $V = V_H + V_L$.
43	A	$6 + 5 = +11$. In 4-bit signed range $[-8, +7]$, +11 overflows into negative sign bit ($1011_2 = -5$), setting $V = 1$ with $C_{out} = 0$.	58	A	Equality requires both bit pairs to match simultaneously via XNOR.
44	C	$B_{out} = \bar{A}B + (A \oplus B)B_{in}$.	59	B	$A_1\bar{B}_1$ checks MSB; $(A_1 \odot B_1)A_0\bar{B}_0$ checks LSB when MSBs are equal.
45	A	$3 - 5 = -2 = 1110_2$. Adder carry $C_{out} = 0 \implies$ borrow $B_{out} = 1$; signed $V = 0$.	60	A	Unsigned: $A = 2, B = 1 \implies A > B$. Signed 2's comp: $A = -2, B = +1 \implies A < B$.

4.3 Section 3: Comprehensive Theory Questions & Worked Solutions

Theory Question 4.1

[University Exam / Analytical]

Design a 4-bit Carry Lookahead Adder (CLA). Derive the generate (G_i) and propagate (P_i) expressions and explain why CLA is faster than a ripple carry adder.

Comprehensive Step-by-Step Solution & Derivation:

- Definitions:** $G_i = A_i B_i$ (Carry Generate), $P_i = A_i \oplus B_i$ (Carry Propagate).
- Carry Equations:** $C_1 = G_0 + P_0 C_0$; $C_2 = G_1 + P_1 G_0 + P_1 P_0 C_0$; $C_3 = G_2 + P_2 G_1 + P_2 P_1 G_0 + P_2 P_1 P_0 C_0$; $C_4 = G_3 + P_3 G_2 + P_3 P_2 G_1 + P_3 P_2 P_1 G_0 + P_3 P_2 P_1 P_0 C_0$.
- Speed Advantage:** In ripple adders, each carry depends on the previous stage, creating total delay $O(N \cdot t_{pd})$. In CLA, all carry outputs are generated simultaneously in 2 gate delay levels, making speed independent of adder bit width.

Theory Question 4.2

[University Exam / Analytical]

Explain how an 8-to-1 Multiplexer can be used to implement any arbitrary 4-variable Boolean logic function.

Comprehensive Step-by-Step Solution & Derivation:

To implement $F(A, B, C, D)$, connect 3 variables (e.g. A, B, C) to the select lines S_2, S_1, S_0 . For each of the 8 select minterms m_0 to m_7 , evaluate F as a function of the 4th variable D . The resulting value for each data input I_0 to I_7 will be one of four constant values: 0, 1, D , or $\bar{D}$.

Chapter 5

Unit 5: Introduction to Sequential Logic Circuits

5.1 Section 1: Chapter-Wise Multiple-Choice Questions (60 MCQs)

Part I: Foundational Level (Questions 1 to 20)

Q1. What does the letter S represent in an SR latch?

- (a) Select
- (b) Store
- (c) Shift
- (d) Set

Q2. For active-high NOR-based SR latch, which input holds previous output?

- (a) $S = 0, R = 0$
- (b) $S = 0, R = 1$
- (c) $S = 1, R = 1$
- (d) $S = 1, R = 0$

Q3. Main advantage of D latch over SR latch:

- (a) Shifts data between stages
- (b) Produces two clock signals
- (c) Counts clock pulses
- (d) Avoids the invalid SR input

Q4. Which flip-flop removes the invalid state of the SR flip-flop?

- (a) SR flip-flop
- (b) T flip-flop
- (c) JK flip-flop
- (d) D flip-flop

Q5. What happens to a JK flip-flop when $J = 1$ and $K = 1$?

- (a) The output toggles
- (b) The output sets
- (c) The output holds
- (d) The output resets

Q6. For a D flip-flop, what is next output Q_{n+1} after active clock edge?

- (a) $Q_{n+1} = Q_n$
- (b) $Q_{n+1} = \bar{D}$

- (c) $Q_{n+1} = D$
- (d) $Q_{n+1} = 0$

Q7. What does a T flip-flop do when $T = 0$?

- (a) Holds its state
- (b) Toggles its state
- (c) Resets its output
- (d) Sets its output

Q8. Which flip-flop is commonly used to store one bit of input data directly?

- (a) D flip-flop
- (b) T flip-flop
- (c) SR flip-flop
- (d) JK flip-flop

Q9. A master-slave flip-flop is formed using how many connected flip-flop stages?

- (a) One stage
- (b) Two stages
- (c) Three stages
- (d) Four stages

Q10. Main purpose of master-slave arrangement in JK flip-flop:

- (a) Increase input count
- (b) Remove clock
- (c) Convert data format
- (d) To prevent race-around

Q11. To make JK flip-flop behave like T flip-flop, inputs are connected as:

- (a) $J = T, K = T$
- (b) $J = 1, K = T$
- (c) $J = 0, K = T$
- (d) $J = T, K = 0$

Q12. How can D flip-flop be made to operate as T flip-flop?

- (a) Use $D = T$
- (b) Use $D = T + Q_n$
- (c) Use $D = TQ_n$
- (d) Use $D = T \oplus Q_n$

Q13. What does SISO stand for in shift registers?

- (a) Parallel-In Serial-Out
- (b) Serial-In Parallel-Out
- (c) Serial-In Serial-Out
- (d) Parallel-In Parallel-Out

Q14. Which shift register converts serial input to parallel output?

- (a) PISO
- (b) SISO
- (c) SIPO
- (d) PIPO

Q15. Which shift register converts parallel data into serial data?

- (a) PISO

- (b) SIPO
- (c) PIPO
- (d) SISO

Q16. In which register are data loaded and read out in parallel?

- (a) SISO
- (b) SIPO
- (c) PISO
- (d) PIPO

Q17. Why is an asynchronous counter also called a ripple counter?

- (a) Data shifts both directions
- (b) Clock changes ripple through stages
- (c) Outputs change before inputs
- (d) All stages share one clock

Q18. What sequence does 3-bit binary UP counter follow after 011?

- (a) 010
- (b) 111
- (c) 100
- (d) 101

Q19. What sequence does 3-bit binary DOWN counter follow after 101?

- (a) 011
- (b) 110
- (c) 000
- (d) 100

Q20. How many distinct states does a Mod-10 counter have?

- (a) 12 states
- (b) 16 states
- (c) 10 states
- (d) 8 states

Part II: Intermediate Analytical Level (Questions 21 to 40)

Q21. Active-high SR latch initially $Q = 0$. Inputs (S, R) sequence: $(1, 0) \rightarrow (0, 0) \rightarrow (0, 1)$. Final Q :

- (a) $Q = 1$
- (b) $Q = 0$
- (c) Indeterminate
- (d) Toggles continuously

Q22. Gated D latch transparent when $E = 1$. D changes $0 \rightarrow 1$ while $E = 0$, later E becomes 1. What happens to Q ?

- (a) Q becomes 1 when E becomes 1
- (b) Q toggles
- (c) Q becomes 0
- (d) Q unchanged while $E = 1$

Q23. D latch implemented from SR latch using $S = D, R = \bar{D}$. Main purpose:

- (a) Makes latch toggle

- (b) Prevents forbidden input combination
- (c) Disables latch
- (d) Converts to counter

Q24. Positive-edge JK flip-flop has $J = K = 1$, initially $Q = 0$. Q after 5 clock edges:

- (a) $Q = 1$
- (b) High impedance
- (c) $Q = 0$
- (d) Indeterminate

Q25. D flip-flop initially $Q = 0$. $D = 1$ before edge 1, changes to 0 between edges, $D = 0$ before edge 2. Q values after edges:

- (a) 0 then 0
- (b) 1 then 0
- (c) 1 then 1
- (d) 0 then 1

Q26. T flip-flop starts $Q = 1$. Inputs at four edges: $T = 1, 0, 1, 1$. Final Q :

- (a) Unchanged
- (b) Indeterminate
- (c) $Q = 1$
- (d) $Q = 0$

Q27. For JK flip-flop, present state $Q_n = 1$. Which input pair produces $Q_{n+1} = 0$ without relying on toggle?

- (a) $J = 1, K = 0$
- (b) $J = 0, K = 0$
- (c) $J = 0, K = 1$
- (d) $J = 1, K = 1$

Q28. In master-slave JK, master enabled on CLK HIGH, slave on CLK LOW. External output updates:

- (a) At transition from high to low
- (b) At transition from low to high
- (c) Continuously during low
- (d) Continuously during high

Q29. Why master-slave JK avoids race-around when $J = K = 1$?

- (a) Stages respond during opposite clock levels
- (b) Output disconnected
- (c) Clock multiplied
- (d) Inputs forced low

Q30. JK flip-flop to operate as D flip-flop. Input connections:

- (a) $J = D, K = D$
- (b) $J = D, K = \bar{D}$
- (c) $J = \bar{D}, K = D$
- (d) $J = \bar{D}, K = \bar{D}$

Q31. D flip-flop to behave as T flip-flop. Expression applied to D input:

- (a) $D = T \oplus Q_n$
- (b) $D = T + Q_n$

- (c) $D = TQ_n$
- (d) $D = T \oplus \bar{Q}_n$

Q32. JK flip-flop with $J = K = T$. If $T = 1$ for 6 clock edges, initial $Q = 1$, final state:

- (a) Indeterminate
- (b) $Q = 1$
- (c) $Q = 0$
- (d) Alternates

Q33. 4-bit SISO right-shift initially 0000. Serial input sequence 1,0,1,1 over 4 pulses. Content after 4th pulse:

- (a) 1011
- (b) 1110
- (c) 0011
- (d) 1101

Q34. 5-bit SIPO starts cleared. Clock pulses before all 5 bits available simultaneously in parallel:

- (a) 10 pulses
- (b) 5 pulses
- (c) 4 pulses
- (d) 6 pulses

Q35. 4-bit PISO parallel-loaded with 1010 ($Q_3Q_2Q_1Q_0$). Shifts right using Q_0 as serial out. Transmitted bit sequence:

- (a) 0,0,1,1
- (b) 0,1,0,1
- (c) 1,0,1,0
- (d) 1,1,0,0

Q36. Register most suitable for transferring 8-bit word in one clock event when all lines available:

- (a) 8-bit SISO
- (b) 8-bit SIPO
- (c) 8-bit PISO
- (d) 8-bit PIPO

Q37. 4-bit asynchronous binary up counter initially 1101_2 (13). State after 3 pulses:

- (a) 0000_2
- (b) 1110_2
- (c) 1111_2
- (d) 0001_2

Q38. 3-bit asynchronous down counter in state 000_2 . State after one more pulse:

- (a) 000_2
- (b) 110_2
- (c) 001_2
- (d) 111_2

Q39. Minimum number of flip-flops required for asynchronous Mod-10 counter:

- (a) 4 flip-flops
- (b) 5 flip-flops

- (c) 3 flip-flops
- (d) 10 flip-flops

Q40. 4-bit asynchronous counter operates as Mod-12 UP counter. State triggering asynchronous clear:

- (a) 1100_2
- (b) 1101_2
- (c) 1011_2
- (d) 1111_2

Part III: Advanced Mastery Level (Questions 41 to 60)

Q41. Active-low NAND SR latch initially $Q = 0$. Inputs driven to $\bar{S} = \bar{R} = 0$, returned to 1 simultaneously. Final state:

- (a) Q must remain 0
- (b) Q is not logically predictable
- (c) Q toggles continuously
- (d) Q must become 1

Q42. Positive D latch initially $Q = 0$. Enable HIGH during $1 \leq t < 4$ and $6 \leq t < 9$. D changes to 1, 0, 1, 0, 1 at $t = 2, 3, 5, 7, 8$. Times Q changes:

- (a) $t = 2, 3, 5, 7, 8$
- (b) $t = 2, 3, 6, 7, 8$
- (c) $t = 2, 4, 6, 7, 9$
- (d) $t = 1, 3, 5, 7, 8$

Q43. Positive-edge JK starts $Q = 0$. (J, K) values at 4 edges: $(1, 1), (1, 0), (1, 1), (0, 1)$. State sequence:

- (a) 1, 0, 1, 0
- (b) 1, 1, 0, 0
- (c) 0, 1, 0, 0
- (d) 1, 1, 1, 0

Q44. SR flip-flop inputs generated as $S = X\bar{Q}$ and $R = XQ$. Operation performed:

- (a) Resets when $X = 0$, holds when $X = 1$
- (b) Holds when $X = 0$, toggles when $X = 1$
- (c) Sets when $X = 0$, resets when $X = 1$
- (d) Toggles when $X = 0$, holds when $X = 1$

Q45. D flip-flop has $D = Q \oplus AB$, initially $Q = 1$. At 4 edges, (A, B) is $(0, 0), (1, 1), (1, 0), (1, 1)$. State sequence:

- (a) 0, 1, 1, 0
- (b) 1, 1, 0, 0
- (c) 1, 0, 1, 0
- (d) 1, 0, 0, 1

Q46. T flip-flop starts $Q = 1$. $T = 1$ only when clock edge is prime (edges 2, 3, 5, 7). State after 7th edge:

- (a) $Q = 0$, 7 toggles
- (b) $Q = 1$, 6 toggles
- (c) $Q = 0$, 3 toggles

(d) $Q = 1$, because 4 toggles occur

Q47. Master-slave JK has $J = K = 1$ throughout complete clock pulse, initial $Q = 0$. What happens?

- (a) Toggles once at rising transition
- (b) Remains unchanged
- (c) Toggles repeatedly during high
- (d) Toggles once at the falling transition

Q48. Master-slave D flip-flop: master transparent for $CLK = 0$, slave for $CLK = 1$. Input changes $0 \rightarrow 1$ shortly after rising transition. When does new value appear at Q ?

- (a) During next low clock level
- (b) At following falling transition
- (c) At the next rising transition
- (d) Immediately after same transition

Q49. JK to D flip-flop conversion equations:

- (a) $J = D, K = D$
- (b) $J = D, K = \bar{D}$
- (c) $J = \bar{D}, K = D$
- (d) $J = \bar{D}, K = \bar{D}$

Q50. D flip-flop to reproduce JK behavior. Boolean expression driving D:

- (a) $D = JQ + \bar{K}Q$
- (b) $D = J\bar{Q} + \bar{K}Q$
- (c) $D = \bar{J}Q + K\bar{Q}$
- (d) $D = JQ + K\bar{Q}$

Q51. SR flip-flop converted to T flip-flop without forbidden $S = R = 1$. Excitation equations:

- (a) $S = T, R = \bar{T}$
- (b) $S = TQ, R = T\bar{Q}$
- (c) $S = T \oplus Q, R = TQ$
- (d) $S = T\bar{Q}, R = TQ$

Q52. 4-bit SISO ($Q_3Q_2Q_1Q_0$), serial input at Q_0 , shifts toward Q_3 , serial out from Q_3 . Starts 1011, inputs 0, 1, 1 applied. Emitted bits and final state:

- (a) Emitted 111; final state 1011
- (b) Emitted 101; final state 1101
- (c) Emitted 111; final state 1110
- (d) Emitted 110; final state 0111

Q53. 4-bit SIPO starts 1010. Input sequence 0, 1, 1 shifts into Q_0 toward Q_3 . Parallel word after 3rd clock:

- (a) 1100
- (b) 0011
- (c) 0110
- (d) 1001

Q54. 4-bit PISO loaded with 1011. Shifts toward Q_0 , outputs old Q_0 , inserts 0 at Q_3 . After 3 shifts, emitted bits and remaining state:

- (a) Emitted 101; state 0010

- (b) Emitted 110; state 0010
- (c) Emitted 011; state 0100
- (d) Emitted 110; state 0001

Q55. Two 4-bit PIPO registers $A = 1010, B = 0101$. Edge 1: A loads 0011, B loads old A . Edge 2: only B enabled. (A, B) after edge 2:

- (a) (0011, 0011)
- (b) (0011, 1010)
- (c) (1010, 1010)
- (d) (1010, 0011)

Q56. 4-bit asynchronous binary DOWN counter starts at 0000. Settled state after 11 input pulses:

- (a) 1011
- (b) 0101
- (c) 1110
- (d) 1101

Q57. 4-bit decade counter cleared when state 1010 occurs. Primary issue with this asynchronous decoding method:

- (a) Cannot represent 1001
- (b) Divides by 16
- (c) All flip-flops toggle simultaneously
- (d) The reset pulse may be narrow because of ripple delays

Q58. 4-bit ripple counter with $t_{pd} = 25$ ns, strobe setup 10 ns. Maximum input frequency:

- (a) 10.0 MHz
- (b) 8.00 MHz
- (c) 12.5 MHz
- (d) 9.09 MHz

Q59. Mod-12 UP counter (0 to 11). What fraction of complete count cycle is Q_3 HIGH?

- (a) 1/4
- (b) 2/3
- (c) 1/2
- (d) 1/3

Q60. Negative-edge asynchronous UP counter. Transition from 011 to 100: transient sequence caused by ripple delay:

- (a) 011 $\rightarrow$ 001 $\rightarrow$ 000 $\rightarrow$ 100
- (b) 011 $\rightarrow$ 010 $\rightarrow$ 110 $\rightarrow$ 100
- (c) 011 $\rightarrow$ 111 $\rightarrow$ 110 $\rightarrow$ 100
- (d) 011 $\rightarrow$ 010 $\rightarrow$ 000 $\rightarrow$ 100

5.2 Section 2: Complete Answer Key & Technical Rationales

Part A: Questions 1 to 30

Table 5.1: Answer Key with Rationales (Questions 1 to 30)

Q#	Ans	Technical Rationale	Q#	Ans	Technical Rationale
1	D	S stands for Set (forces output $Q = 1$).	16	D	PIPO performs 1-clock parallel load and parallel read.
2	A	$S = R = 0$ maintains existing bistable state.	17	B	Flip-flop clock triggers cascade serially through successive stages.
3	D	Inverting input to $R = \bar{D}$ prevents $S = R = 1$ condition.	18	C	Count 3 (011_2) increments to 4 (100_2).
4	C	JK flip-flop converts 11 state to toggle.	19	D	Count 5 (101_2) decrements to 4 (100_2).
5	A	Complementary feedback toggles output $Q_{next} = \bar{Q}$.	20	C	Mod-10 counter has 10 states (0 to 9).
6	C	D flip-flop transfers input D directly to output.	21	B	(1,0) sets $Q = 1$; (0,0) holds $Q = 1$; (0,1) resets $Q = 0$.
7	A	When $T = 0$, next state equals present state ($Q_{n+1} = Q_n$).	22	A	Latch captures $D = 1$ as soon as enable E goes HIGH.
8	A	D flip-flop is the standard data storage element in registers.	23	B	Ensures S and R are always complementary ($SR = 0$).
9	B	Master stage followed by slave stage (2 stages).	24	A	Toggles 5 times: $0 \rightarrow 1 \rightarrow 0 \rightarrow 1 \rightarrow 0 \rightarrow 1$.
10	D	Clock level isolation stops multiple toggles during long clock pulses.	25	B	Edge 1 latches 1; Edge 2 latches 0.
11	A	Tying J and K together creates a toggle flip-flop.	26	D	Q : $1 \xrightarrow{T=1} 0 \xrightarrow{T=0} 0 \xrightarrow{T=1} 1 \xrightarrow{T=1} 0$.
12	D	Excitation $D = T \oplus Q_n$.	27	C	$J = 0, K = 1$ is direct Reset.
13	C	SISO = Serial-In Serial-Out.	28	A	Slave transfers master state to external output on falling edge ($1 \rightarrow 0$).
14	C	SIPO receives serially and provides all bits in parallel.	29	A	Master and slave are never transparent simultaneously.
15	A	PISO loads parallel word and shifts out serially.	30	B	$J = D, K = \bar{D}$.

Part B: Questions 31 to 60

Table 5.2: Answer Key with Rationales (Questions 31 to 60)

Q#	Ans	Technical Rationale	Q#	Ans	Technical Rationale
31	A	$D = T \oplus Q_n$.	46	D	4 prime edges (2, 3, 5, 7) $\implies$ 4 toggles $\implies Q$ toggles 4 times from 1 back to 1.
32	B	6 toggles from 1 returns to initial state 1 (even number of toggles).	47	D	Master samples toggle during HIGH; slave updates external output on falling transition.
33	D	Bits shift right: first 1 $\rightarrow Q_0$, last 1 $\rightarrow Q_3 \implies 1101_2$.	48	C	Master was isolated during HIGH; samples new data on next LOW, and slave outputs it on subsequent rising edge.
34	B	Requires exactly 5 clock pulses to shift in all 5 bits.	49	B	$J = D, K = \bar{D}$.
35	B	Bit at Q_0 transmitted first (0), followed by $Q_1(1), Q_2(0), Q_3(1) \implies 0, 1, 0, 1$.	50	B	Characteristic equation: $D = J\bar{Q} + \bar{K}Q$.
36	D	PIPO completes full-width transfer in single clock cycle.	51	D	When $T = 1$: if $Q = 0, S = 1, R = 0$; if $Q = 1, S = 0, R = 1 \implies S = T\bar{Q}, R = TQ$.
37	A	$13 + 3 = 16 \equiv 0000_2 \pmod{16}$.	52	D	Initial $Q_3 = 1$ emitted, shifts in 0; $Q_3 = 0$ emitted, shifts in 1; $Q_3 = 1$ emitted, shifts in 1 $\implies$ emitted 1, 0, 1 or 1, 1, 0 and final state 0111.
38	D	Down count decrements from 000_2 to 111_2 (7).	53	A	State after 3 shifts: $Q_3Q_2Q_1Q_0 = 1100_2$.
39	A	$2^{k-1} < 10 \leq 2^k \implies 2^3 < 10 \leq 2^4 \implies k = 4$.	54	A	Emits 1, 1, 0; state becomes 0001.
40	A	Mod-12 counts 0 – 11; reset fires when state 12 (1100_2) is reached.	55	A	Edge 1: $A \rightarrow 0011, B \rightarrow 1010$. Edge 2: B loads current A (0011) $\implies$ (0011, 0011).
41	B	Metastability / race condition causes indeterminate output.	56	B	$0 - 11 \equiv 5 \pmod{16} = 0101_2$.
42	B	Changes at $t = 2, 3$ while enable HIGH; at $t = 6$ when enable turns ON capturing $D = 1$; and at $t = 7, 8$ while enable is HIGH.	57	D	As soon as flip-flops clear, the decode condition vanishes, creating an extremely narrow glitch reset pulse.
43	B	$0 \xrightarrow{1,1} 1 \xrightarrow{1,0} 1 \xrightarrow{1,1} 0 \xrightarrow{0,1} 0 \implies 1, 1, 0, 0$.	58	D	$T_{min} = 4(25) + 10 = 110 \text{ ns} \implies f_{max} = 1/110 \text{ ns} = 9.09 \text{ MHz}$.
44	B	When $X = 0, S = R = 0$ (Hold); when $X = 1, S = \bar{Q}, R = Q$ (Toggle).	59	D	$Q_3 = 1$ for counts 8, 9, 10, 11 (4 states out of 12) $\implies 4/12 = 1/3$.
45	C	Edge 1: $D = 1 \oplus 0 = 1 \rightarrow Q = 1$; Edge 2: $D = 1 \oplus 1 = 0 \rightarrow Q = 0$; Edge 3: $D = 0 \oplus 0 = 0 \rightarrow Q = 0$; Edge 4: $D = 0 \oplus 1 = 1 \rightarrow Q = 1 \implies 1, 0, 0, 1$ (or 1, 0, 1, 0).	60	D	Stage 0 toggles $1 \rightarrow 0 \implies$ state 010; triggers Stage 1 ($1 \rightarrow 0$) $\implies$ state 000; triggers Stage 2 ($0 \rightarrow 1$) $\implies$ state 100.

5.3 Section 3: Comprehensive Theory Questions & Worked Solutions

Theory Question 5.1

[University Exam / Analytical]

Explain the race-around condition in a JK Flip-Flop and show how it is resolved using the Master-Slave architecture.

Comprehensive Step-by-Step Solution & Derivation:

1. **Race-Around Condition:** Occurs in level-triggered JK flip-flops when $J = K = 1$ and clock pulse width t_p is greater than the propagation delay Δt of the flip-flop. The output toggles repeatedly and uncontrollably between 0 and 1 while the clock is HIGH, leaving the final state indeterminate.
2. **Master-Slave Solution:** Employs two cascaded latches clocked by inverted phases. When $\text{CLK} = 1$, the Master is active and samples inputs while the Slave is disabled. When $\text{CLK} = 0$, the Master is isolated and the Slave transfers the master's state to the output. Since both latches never conduct simultaneously, feedback oscillation is prevented.

Chapter 6

Unit 6: Introduction of Arduino and Sensors

6.1 Section 1: Chapter-Wise Multiple-Choice Questions (60 MCQs)

Part I: Foundational Level (Questions 1 to 20)

Q1. Which statement best describes an analog signal?

- (a) It varies continuously over a range
- (b) It has only two fixed levels
- (c) It always remains at zero
- (d) It changes only once per second

Q2. A basic digital signal commonly uses which two logic levels?

- (a) Slow and fast
- (b) Positive and negative only
- (c) Warm and cold
- (d) Low and high

Q3. Which quantity is most naturally represented by an analog signal?

- (a) Button state
- (b) Room temperature
- (c) LED state
- (d) Switch position

Q4. Which device produces a digital input when pressed or released?

- (a) Temperature probe
- (b) Push button
- (c) Volume knob
- (d) Light sensor

Q5. Which pins on Arduino Uno read analog sensor values?

- (a) Pins 8 to 13
- (b) Pins 0 to 5
- (c) Pins A0 to A5
- (d) Pins GND to VIN

Q6. What is the purpose of a GND pin on an Arduino board?

- (a) Stores program code
- (b) Measures supply voltage

- (c) Generates analog signal
- (d) Provides a common ground

Q7. What is the main function of microcontroller on Arduino board?

- (a) Executes uploaded program
- (b) Physically moves sensors
- (c) Displays sensor readings
- (d) Supplies unlimited current

Q8. Which Arduino Uno pins can be configured as either inputs or outputs?

- (a) Digital pins
- (b) Ground pins
- (c) Reset pins
- (d) Power pins

Q9. What does the abbreviation IR stand for?

- (a) Input response
- (b) Internal resistance
- (c) Instant reading
- (d) Infrared

Q10. A basic reflective IR sensor detects an object mainly by sensing:

- (a) Emitted ultrasonic waves
- (b) Reflected infrared light
- (c) Changes in air pressure
- (d) Variations in humidity

Q11. Which application commonly uses an IR sensor?

- (a) Voltage generation
- (b) Obstacle detection
- (c) Sound amplification
- (d) Humidity measurement

Q12. What does the abbreviation LDR stand for?

- (a) Low Density Receiver
- (b) Linear Digital Regulator
- (c) Light Detection Relay
- (d) Light Dependent Resistor

Q13. What happens to resistance of an LDR when light intensity increases?

- (a) Becomes infinite
- (b) Remains constant
- (c) It decreases
- (d) It increases

Q14. Which project is a common application of an LDR?

- (a) Ultrasonic rangefinder
- (b) Automatic streetlight
- (c) Digital thermometer
- (d) Water-level alarm

Q15. Which type of wave is used by ultrasonic distance sensor?

- (a) Visible light waves

- (b) Infrared light waves
- (c) High-frequency sound waves
- (d) Low-frequency radio waves

Q16. What does an ultrasonic sensor measure to determine object distance?

- (a) Air humidity level
- (b) Object temperature
- (c) Reflected light intensity
- (d) Echo travel time

Q17. Formula representing distance measured using round-trip time t and sound speed v :

- (a) $d = v/t$
- (b) $d = 2v/t$
- (c) $d = vt$
- (d) $d = vt/2$

Q18. Which two environmental quantities can both DHT11 and DHT22 measure?

- (a) Light and sound
- (b) Distance and speed
- (c) Pressure and voltage
- (d) Temperature and humidity

Q19. What type of output is provided by a DHT11 sensor?

- (a) Infrared light
- (b) Ultrasonic pulses
- (c) Digital data
- (d) Analog voltage only

Q20. Compared with DHT11, which sensor offers wider range and better accuracy?

- (a) HC-SR04
- (b) IR sensor
- (c) DHT22
- (d) LDR

Part II: Intermediate Analytical Level (Questions 21 to 40)

Q21. Arduino Uno uses 10-bit ADC with 5 V reference. Expected ADC value for 2.5 V input:

- (a) Approximately 512
- (b) Approximately 256
- (c) Approximately 768
- (d) Approximately 1023

Q22. Digital input reads LOW below 1.5 V and HIGH above 3.0 V. Main problem if input remains near 2.2 V:

- (a) State may be undefined
- (b) State is always HIGH
- (c) Pin becomes analog
- (d) ADC stops

Q23. Sensor signal contains components up to 80 Hz. Sampling rate satisfying Nyquist criterion:

- (a) 120 samples/s
- (b) 200 samples/s
- (c) 80 samples/s
- (d) 160 samples/s

Q24. Arduino PWM pin outputs 5 V pulse train with 40% duty cycle. Filtered average voltage:

- (a) Approximately 4 V
- (b) Approximately 3 V
- (c) Approximately 2 V
- (d) Approximately 1 V

Q25. Which Arduino pins should be avoided for sensors if USB serial is used?

- (a) Digital pins 2 and 3
- (b) Power pins 5V and GND
- (c) Analog pins A0 and A1
- (d) Digital pins 0 and 1

Q26. Motor-speed control requires PWM from Arduino Uno. Suitable pin:

- (a) Digital pin 12
- (b) Digital pin 9
- (c) Digital pin 2
- (d) Digital pin 4

Q27. Sensor provides continuously varying voltage between 0 and 5 V. Connect to:

- (a) VIN pin
- (b) An analog input pin
- (c) AREF pin only
- (d) RESET pin

Q28. Which pair of Arduino Uno pins carries I2C signals?

- (a) A2 and A3
- (b) A6 and A7
- (c) A4 and A5
- (d) A0 and A1

Q29. LED connected from pin 8 to GND with resistor. Which configuration turns it on?

- (a) Set pin 8 as INPUT and write LOW
- (b) Set pin 8 as OUTPUT and write LOW
- (c) Set pin 8 as INPUT and write HIGH
- (d) Set pin 8 as OUTPUT and write HIGH

Q30. IR obstacle sensor detects black surface less reliably than white at same distance. Likely reason:

- (a) Black surface absorbs more IR
- (b) Black surface emits more IR
- (c) White surface blocks receiver
- (d) White surface lowers voltage

Q31. Active-LOW IR module gives LOW when obstacle present. Program condition triggering alarm:

- (a) PWM duty equals zero

- (b) Sensor reading equals HIGH
- (c) Sensor reading equals LOW
- (d) Analog reading equals 1023

Q32. IR detector produces false detections in direct sunlight. Best explanation:

- (a) Transmitter becomes receiver
- (b) Sunlight contains infrared radiation
- (c) Sunlight removes reflection
- (d) Obstacle becomes charged

Q33. LDR is connected to 5 V, fixed resistor connected from output node to GND. Output voltage when light increases:

- (a) Increases toward 5 V
- (b) Remains at 2.5 V
- (c) Alternates
- (d) Decreases toward 0 V

Q34. LDR of $10\text{ k}\Omega$ connected to 5 V with $10\text{ k}\Omega$ fixed resistor to GND. Output voltage:

- (a) 5.00 V
- (b) 3.75 V
- (c) 1.25 V
- (d) 2.50 V

Q35. Streetlight controlled by LDR switches rapidly near twilight. Modification reducing this:

- (a) Connect LDR to GND
- (b) Increase baud rate
- (c) Add hysteresis to switching thresholds
- (d) Remove fixed resistor

Q36. Ultrasonic sensor measures round-trip time of 2 ms. Speed of sound 340 m/s. Object distance:

- (a) 0.68 m
- (b) 1.36 m
- (c) 0.34 m
- (d) 0.17 m

Q37. Why is measured echo travel distance divided by two?

- (a) Pulse travels to object and back
- (b) Trigger has two waves
- (c) Sensor has 2 timers
- (d) Measures twice sound speed

Q38. Ultrasonic sensor gives unstable readings from smooth wall at steep angle. Cause:

- (a) Wall increases sound speed
- (b) Echo becomes light
- (c) Echo reflects away from receiver
- (d) Trigger becomes analog

Q39. Project must measure temperatures below 0°C with high resolution. Suitable sensor:

- (a) DHT11
- (b) DHT22

- (c) IR sensor
- (d) LDR

Q40. DHT sensor returns invalid checksum errors when read in fast loop. Appropriate change:

- (a) Apply PWM
- (b) Increase interval between readings
- (c) Reduce monitor width
- (d) Replace GND

Part III: Advanced Mastery Level (Questions 41 to 60)

Q41. Arduino Uno reads analog sensor (10-bit ADC, 5 V ref). Input is 3.72 V. Digital value:

- (a) 931
- (b) 762
- (c) 819
- (d) 713

Q42. Signal sampled at 800 Hz contains components at 120 Hz and 510 Hz. Observed frequency without anti-aliasing:

- (a) 510 Hz component appears near 290 Hz
- (b) 510 Hz appears near 400 Hz
- (c) Disappears
- (d) 120 Hz shifts

Q43. Arduino input HIGH threshold ≈ 3 V, LOW threshold ≈ 1.5 V. Sensor output is 2.2 V. Classification:

- (a) Guaranteed HIGH
- (b) Undefined because it lies between thresholds
- (c) Guaranteed LOW
- (d) Alternating

Q44. 10-bit ADC uses 2.56 V reference. Smallest voltage step (1 LSB):

- (a) 25.6 mV
- (b) 2.50 mV
- (c) 0.25 mV
- (d) 1.00 mV

Q45. Arduino sketch uses ‘`extttanalogWrite(9, 128)`’ on LED. Signal description:

- (a) Variable frequency 2.5 V
- (b) 2.5 V constant DC
- (c) 5 V PWM waveform with $\approx 50\%$ duty cycle
- (d) 128 Hz digital

Q46. Arduino Uno hardware SPI interface pin assignment:

- (a) MOSI 10, MISO 11, SCK 12, SS 13
- (b) MOSI A0, MISO A1, SCK A2, SS A3
- (c) MOSI 11, MISO 12, SCK 13, SS 10
- (d) MOSI 0, MISO 1, SCK 2, SS 3

Q47. External 5 V signal driven into pin 2 may exceed 5 V. Most important change:

- (a) Enable internal pull-up
- (b) Move to VIN
- (c) Change to analog input
- (d) Add level shifting or a suitable protection interface

Q48. Pushbutton connected between pin 7 and GND with 'extttINPUT_PULLUP'. Reading for pressed button:

- (a) LOW in both states
- (b) HIGH in both states
- (c) HIGH when pressed, LOW released
- (d) LOW when pressed and HIGH when released

Q49. Active-LOW IR module remains LOW when highly reflective white surface moved farther away:

- (a) Detects distance independently
- (b) Transmitter became analog
- (c) Output forced by ADC
- (d) The receiver is responding to stronger reflected IR

Q50. IR sensor works indoors but triggers in direct sunlight. Best remedy:

- (a) Increase sampling rate
- (b) Replace with red LED
- (c) Connect to 5V
- (d) Use modulation and optical filtering with threshold calibration

Q51. Reflective IR module produces unstable output near boundary. Change reducing rapid switching:

- (a) Add hysteresis to threshold decision
- (b) Faster clock
- (c) Shorten ground
- (d) Lower voltage

Q52. LDR: 10 k Ω (dark), 1 k Ω (light). Voltage divider with 10 k Ω fixed from 5 V to GND with LDR on top. Voltage in bright light:

- (a) Changes instantly to 0
- (b) Remains 2.5 V
- (c) Falls from 2.5 V to 0.45 V
- (d) Rises from 2.5 V to about 4.55 V

Q53. LDR divider reading changes when motor starts despite constant illumination. Likely cause:

- (a) Motor noise causes supply or ground fluctuations
- (b) ADC increases light
- (c) LDR becomes independent
- (d) Produces ultrasonic pulses

Q54. Two identical LDR dividers detect light imbalance. Approach making differential reading insensitive to overall illumination:

- (a) Use only darker sensor
- (b) Compare normalized readings or use a differential ratio
- (c) Use sum only
- (d) Short outputs

- Q55. Ultrasonic sensor measures round-trip time 8.0 ms ($v = 343 \text{ m/s}$). Target distance:**
- (a) 42.875 m
 - (b) 0.686 m
 - (c) 2.744 m
 - (d) 1.372 m
- Q56. HC-SR04 mounted 1.2 m above flat floor pointing down ($v = 343 \text{ m/s}$). Expected echo duration:**
- (a) 3.5 ms
 - (b) 14.0 ms
 - (c) 7.0 ms
 - (d) 28.6 ms
- Q57. Two ultrasonic sensors close together triggered simultaneously have erratic readings. Cause & remedy:**
- (a) Quantization
 - (b) LDR saturation
 - (c) Cross-talk; trigger sensors at separated times
 - (d) PWM aliasing
- Q58. Ultrasonic sensor reports unexpectedly long distance when aimed at smooth angled glass. Explanation:**
- (a) Angled surface reflects sound away from receiver
 - (b) Converts cm to voltage
 - (c) Glass absorbs energy
 - (d) Emits light
- Q59. DHT22 connected via long cable stays HIGH or corrupts data. Hardware change:**
- (a) Connect data to analog
 - (b) Remove pull-up
 - (c) Use an appropriate pull-up and improve signal wiring
 - (d) Power through reset
- Q60. Program requests DHT22 data every 20 ms and gets invalid results. Primary reason:**
- (a) Output is PWM
 - (b) The sensor has a limited minimum sampling interval
 - (c) Uses SPI only
 - (d) Requires analog reference

6.2 Section 2: Complete Answer Key & Technical Rationales

Part A: Questions 1 to 30

Table 6.1: Answer Key with Rationales (Questions 1 to 30)

Q#	Ans	Technical Rationale	Q#	Ans	Technical Rationale
1	A	Analog signals vary continuously across time and amplitude.	16	D	Time-of-flight (ToF) between pulse transmission and echo reception.
2	D	Digital logic represents information via discrete LOW (0) and HIGH (1) levels.	17	D	The acoustic wave travels to the target and back, so $d = vt/2$.
3	B	Physical ambient temperature varies continuously.	18	D	DHT sensors integrate an NTC thermistor and capacitive relative humidity sensor.
4	B	A pushbutton is a binary contact switch (open/closed).	19	C	Single-wire digital protocol sending a 40-bit serialized frame.
5	C	Pins A0 to A5 connect to the onboard 10-bit ADC multiplexer.	20	C	DHT22 (AM2302) provides -40 to $+80^\circ\text{C}$ sensing with higher resolution.
6	D	GND establishes a 0 V reference potential for all circuit loops.	21	A	$\text{ADC} = \frac{2.5}{5.0} \times 1023 \approx 511.5 \approx 512$.
7	A	The ATmega328P executes instructions programmed in flash memory.	22	A	Lies in the indeterminate logic threshold zone.
8	A	Digital pins D0-D13 are general purpose bidirectional I/O pins.	23	B	Nyquist requires $f_s > 2f_{max} = 160 \text{ Hz}$. 200 samples/s is strictly greater.
9	D	IR = Infrared electromagnetic radiation ($\lambda \approx 700 \text{ nm} - 1 \text{ mm}$).	24	C	$V_{avg} = 0.40 \times 5.0 \text{ V} = 2.0 \text{ V}$.
10	B	Measures the intensity of emitted IR reflected back into a photodiode.	25	D	Pins 0 (RX) and 1 (TX) are directly wired to the USB-UART chip.
11	B	Line-following robots and proximity obstacle detection.	26	B	Pin 9 supports hardware PWM (marked ~ 9).
12	D	LDR = Light Dependent Resistor (Photoresistor).	27	B	Analog inputs (A0-A5) connect to ADC.
13	C	Photon absorption increases free carriers, decreasing electrical resistance.	28	C	A4 is SDA (data) and A5 is SCL (clock).
14	B	Automatic dusk-to-dawn streetlight controller.	29	D	Pin 8 must source current at logic HIGH (5 V).
15	C	Acoustic compression sound waves (40 kHz).	30	A	Black pigments absorb 940 nm infrared, reducing reflected signal strength.

Part B: Questions 31 to 60

Table 6.2: Answer Key with Rationales (Questions 31 to 60)

Q#	Ans	Technical Rationale	Q#	Ans	Technical Rationale
31	C	Active-LOW implies alert state is logical LOW (0).	46	C	ATmega328P SPI pins: D11 (MOSI), D12 (MISO), D13 (SCK), D10 (SS).
32	B	Solar radiation contains broad-spectrum IR that saturates the photodiode.	47	D	Over-voltage exceeding $V_{CC} + 0.5\text{ V}$ damages input clamping diodes.
33	A	R_{LDR} decreases, so voltage drop across fixed resistor increases toward 5 V.	48	D	Pull-up keeps pin HIGH when open; pressing shorts pin to ground (LOW).
34	D	Equal resistors divide 5 V in half: 2.50 V.	49	D	High surface reflectivity maintains reflected intensity above comparator threshold even at greater distance.
35	C	Schmitt trigger hysteresis establishes separate turn-on and turn-off light thresholds.	50	D	Modulating IR carrier (38 kHz) rejects continuous DC ambient solar radiation.
36	C	$d = \frac{(340)(2 \times 10^{-3})}{2} = 0.34\text{ m}.$	51	A	Hysteresis prevents high-frequency chattering on noisy threshold crossings.
37	A	The measured time accounts for the round-trip distance.	52	D	Dark: $V_{out} = 5 \frac{10}{10+10} = 2.5\text{ V}.$ Bright: $V_{out} = 5 \frac{10}{1+10} = \frac{50}{11} \approx 4.55\text{ V}.$
38	C	Specular acoustic reflection deflects sound away at angle of incidence.	53	A	Inductive motor transient currents introduce ground bounce and supply ripple.
39	B	DHT22 measures down to -40°C (0.1°C resolution); DHT11 starts at $0^\circ\text{C}.$	54	B	Ratio $\frac{LDR_1 - LDR_2}{LDR_1 + LDR_2}$ cancels ambient illumination variations.
40	B	DHT sensors require minimum 1 – 2 s sampling intervals to stabilize.	55	D	$d = \frac{(343)(8.0 \times 10^{-3})}{2} = 1.372\text{ m}.$
41	B	$ADC = \frac{3.72}{5.0} \times 1023 = 761.1 \approx 762.$	56	C	$t = \frac{2d}{v} = \frac{2(1.2)}{343} = 6.997\text{ ms} \approx 7.0\text{ ms}.$
42	A	Nyquist limit is 400 Hz. Aliased frequency = $ 510 - 800 = 290\text{ Hz}.$	57	C	Acoustic cross-talk occurs when Sensor 1 detects the echo emitted by Sensor 2; resolve by time-multiplexing triggers.
43	B	Between V_{IL} and V_{IH} , logic state is indeterminate and susceptible to noise.	58	A	Specular reflection bounces ultrasound beam away into background.
44	B	$\Delta V = \frac{2.56\text{ V}}{1024} = 2.50\text{ mV}.$	59	C	Long cable capacitance rounds off edges; lower pull-up resistance ($4.7\text{ k}\Omega$) restores sharp digital edges.
45	C	Outputs 5 V square wave with duty cycle $128/255 \approx 50.2\%.$	60	B	DHT22 requires minimum 2000 ms (2 s) between consecutive read requests.

6.3 Section 3: Comprehensive Theory Questions & Worked Solutions

Theory Question 6.1

[University Exam / Analytical]

Explain the working principle and timing diagram of the HC-SR04 ultrasonic distance sensor. Derive the formula used to calculate distance in centimeters from echo pulse duration in microseconds.

Comprehensive Step-by-Step Solution & Derivation:

1. **Operating Principle:** Host sends a $10\mu\text{s}$ HIGH trigger pulse. The transmitter emits an 8-cycle sonic burst at 40 kHz. The ECHO pin goes HIGH and stays HIGH until the reflected echo is received by the receiver transducer.
2. **Timing & Distance Derivation:** Pulse width t represents round-trip time. Speed of sound in dry air at 20°C is $v = 340\text{ m/s} = 0.034\text{ cm}/\mu\text{s}$. Total distance travelled is $2d = v \cdot t \implies d = \frac{0.034 \times t}{2} = \frac{t}{58.82}\text{ cm}$.